How Does Privacy Regulation Affect Transatlantic Venture Investment? Evidence from GDPR*

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Abstract

We examine how privacy regulation affects the allocation and organization of cross-border venture financing. Using 2014–2019 transaction and investor data, a triple-difference design compares more- and less-data-exposed industries in Europe and the United States around the introduction of the GDPR. The policy rollout leads to a 19.2 percent relative decline in financing rounds involving US investors without EU participation in exposed European industries. Following established investors, we also find reduced US participation as syndicate leads and solo investors, conditional on their quarterly activity in these roles. Lead-investor distances shorten among completed rounds. These patterns suggest that the financing implications of privacy regulation extend to foreign investors' involvement in evaluating ventures and organizing their financing. The findings highlight access to foreign investment leadership as a relevant consideration alongside access to capital in an evolving regulatory environment.

Keywords: GDPR, privacy, venture capital, technology venture, local preference, syndication.

JEL Codes: G11, K20, D8.

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1. Introduction

Venture capitalists play a crucial role in driving economic growth by providing financial resources to young and innovative companies, typically in exchange for equity stakes (Kortum and Lerner 2000; Samila and Sorenson 2011). The United States has the largest and most developed venture capital (VC) market worldwide, with Europe lagging far behind. During the past decade, annual VC investments in the EU averaged 0.2% of GDP, less than one-third of the 0.7% observed in the US (Arnold, Claveres and Frie 2024). Although this lower volume of venture capital investment may, in principle, reflect a relative shortage of successful, high-growth potential ventures on the demand side, it underscores persistent supply-side issues. Specifically, the EU has fewer large-scale VC funds, and those that exist are often less equipped than their US counterparts. Since 2013, EU-based VC funds raised approximately $794 billion less than their US counterparts and there have been 137 VC funds larger than USD 1 billion in the US, compared to only 11 in the EU.[1] This gap highlights the importance of cross-border investment inflows, particularly from US investors, into economies characterized by less developed venture investment ecosystems. Moreover, foreign investors can bring valuable expertise, facilitating the internationalization of local ventures and opening opportunities for foreign exits (Mäkelä and Maula 2005; Jääskeläinen and Maula 2005; Hursti and Maula 2007), especially when foreign venture capitalists (VCs) partner with local ones (Schertler and Tykvová 2011; Humphery-Jenner and Suchard 2013). In effect, according to the 2023 European Investment Fund Venture Capital Survey,[2] over 80% of VC fund managers agree that greater activity by global investors is crucial for scale-up financing in Europe.

Against this backdrop, data has become a key input in technology, attracting investments in data-driven innovations around the world. Such innovations have transformed

[1] See Arnold, Claveres and Frie (2024) and Draghi (2024) for additional details.

[2] Available at https://www.eif.org/news_centre/publications/eif_working_paper_2023_93.pdf, accessed on 1/14/2025.

marketplaces, and the prospects of new services and monetization surrounding data have galvanized entrepreneurs and investors across industry sectors. However, data-driven operations are also associated with concerns about exposure, privacy intrusion, and misuse without the knowledge or consent of the source. These concerns, exacerbated by recent cases of data breaches and related scandals, have led to calls for tighter rulesets that require more transparency, control, and the imposition of some limits on the collection, storage, and processing of users' data.[3] However, adding regulatory burdens to companies can hurt innovation and entrepreneurship (Djankov et al. 2002; Bleier, Goldfarb and Tucker 2020). This negative effect can be particularly large among smaller technology ventures seeking to raise venture capital, a context where shifting regulatory landscapes have been shown to contribute to raising uncertainty in returns and potentially lowering investment (Gompers and Lerner 2001). Consistently, more than half of the small and medium enterprises in Europe flag regulatory obstacles and the administrative burden as their greatest challenge.[4]

The EU's relatively underdeveloped venture investment ecosystem, together with the regulatory burdens on young and innovative European technology ventures, have been identified as key drivers of the innovation and competitiveness gaps with the US by Draghi (2024). In this paper, we examine the unintended consequences of the European General Data Protection Regulation (GDPR) in affecting investment flows between the EU and US.

The GDPR, a landmark EU privacy law that imposes conditions on firms' data

[3]See for example the reports published by the Federal Trade Commission in 2012 (available at: https://www.ftc.gov/reports/protecting-consumer-privacy-era-rapid-change-recommendations-businesses-policymakers), 2014 (available at: https://www.ftc.gov/reports/data-brokers-call-transparency-accountability-report-federal-trade-commission-may-2014) and 2016 (available at: https://www.ftc.gov/reports/big-data-tool-inclusion-or-exclusion-understanding-issues-ftc-report), or by the European Data Protection Supervisor in 2016 (available at: https://www.edps.europa.eu/sites/default/files/publication/16-09-23_bigdata_opinion_en.pdf).

[4]See for example BusinessEurope's report, "Reducing Regulatory Burden to Restore the EU's Competitive Edge," available at: https://www.businesseurope.eu/publications/reducing-regulatory-burden-restore-eus-competitive-edge?.

practices, was enacted on April 27, 2016, becoming enforceable two years later on May 25, 2018. The regulation aims to protect data by 'design and default,' with both specific and heuristic requirements that firms handle data according to a set of principles. For the youngest of those firms and their investors, GDPR introduces significant relative costs and uncertainty. First, the regulation creates uncertainty with respect to which data-driven products are compliant and whether products or processes need to be changed because compliance itself is a function of heuristics that have yet to be fully tested in the courts.[5] Second, ventures may rely on the compliance strategies of larger platforms, but many of these platforms only announced how they intended to pursue compliance on or around GDPR's implementation date,[6] and some have, over time, revised their policies to reach compliance adequacy.[7] The choices of the larger platforms may also be critical for smaller firms because they influence those firms' data-related liabilities under the regulation.[8] As a consequence, the actual cost of compliance may change over time. The regulation is therefore associated with uncertainty about the extent to which ventures can get their products to comply and how much it would cost them to do so, with early indications that the costs may be significant (Goldberg, Johnson and Shriver 2024). For investors, it follows that GDPR introduces new uncertainty, as well as

[5]For instance, ventures and investors may be unclear about whether legitimate interest (versus informed consent) is an adequate path to compliance. See, e.g., https://www.cpomagazine.com/data-protection/direct-marketing-under-the-gdpr-consent-vs-legitimate-interests/ and https://www.assetfinanceinternational.com/index.php/legal/legal-general/legal-general/17331-gdpr-confusion-over-legitimate-interest-affects-motor-retail-marketing.

[6]Examples include a SafeDK report that more than half of mobile applications are not compliant (https://www.mobilemarketer.com/news/study-55-of-apps-may-not-meet-gdpr-privacy-standards/515546/), Apple reportedly removing apps that share location data (https://www.idownloadblog.com/2018/05/09/apple-removing-apps-location-data/) and updating its privacy terms (https://techcrunch.com/2018/05/23/apple-introduces-new-privacy-portal-to-comply-with-gdpr/), Facebook announcing that "Businesses may want to implement code that creates a banner and requires affirmative consent. Each company is responsible for ensuring their own compliance" (https://developers.facebook.com/ads/blog/post/2018/05/10/compliance-protections-gdpr/), Shopify updating its app permissions for merchants and developers (https://www.shopify.com/partners/blog/gdpr-compliance), and Google releasing new consent requirements to developers (https://bit.ly/2ziUgJA).

[7]See, e.g., https://www.nytimes.com/2019/01/21/technology/google-europe-gdpr-fine.html and https://www.ft.com/content/197a6758-a148-11e9-a282-2df48f366f7d.

[8]See, e.g., https://digiday.com/media/google-publishers-gdpr-standards/.

information acquisition and due diligence costs related to EU venture deals, and these costs may be particularly pronounced for investors who are not based in the EU.

In this paper, we leverage the introduction of the GDPR as an exogenous variation. Using six years of investment data (2014-2019), we investigate the impact of the GDPR on investors' investment strategies and its implications for venture investment flows between the EU and US. Common strategies that venture investors may adopt to mitigate the increased investment costs following a law change include investing locally and syndicating deals (e.g., Gompers and Lerner 1999; Cumming and Dai 2010). Accordingly, we conjecture that US investors could reduce their investments in EU ventures operating in industries affected by the regulation and shift their focus to other unaffected assets a strategy to which we refer as "pullback."

Using a triple-difference design that compares more- and less-data-exposed industries in the EU and US, we find a relative contraction in foreign financing for exposed European industries during the GDPR's rollout. The number of rounds involving US investors without EU participation declines by 19.2%, while rounds led by US investors decline by 27.8%. The corresponding estimates for rounds only involving or led by EU investors are close to zero. These results indicate that exposed European industries experienced a relative loss of foreign financing even as less-exposed European industries continued to attract US investment.

Following individual investors active before the GDPR's enactment, we find a 12.6% relative decline in US investor participation in exposed European industries. This specification allows the response to reflect changes in both investors' overall activity and its allocation across destinations. Adding investor-by-quarter fixed effects yields a similar decline of 11.8%. This evidence points to a role played by changes in capital allocation among existing investors in explaining the reduced aggregate inflow of transatlantic investment in Europe. The clearest investor-level declines occur in roles that entail primary responsibility for an investment. We find that US participation as a syndicate lead

falls by 32.1%, and participation as a solo investor falls by 29.4%. The corresponding estimate for participation as a syndicate non-lead is positive but statistically insignificant. Completed investments also become geographically more local: average lead-investor distance declines by 25.2%, while the non-lead distance estimate is close to zero. Together, these patterns are consistent with regulatory exposure increasing the importance of local information and expertise in screening and monitoring ventures.

Overall, our findings suggest that privacy regulation can shape both the allocation of entrepreneurial financing and the roles foreign investors undertake. The distinction matters because foreign investors can contribute expertise, networks, and access to international markets alongside capital. A relative reduction in their participation as lead or solo investors may therefore affect the financial intermediation available to European ventures beyond the amounts invested. From a policy perspective, the results motivate evaluating how the implementation of privacy regulation affects access to foreign investment leadership, alongside its effects on financing volumes and its intended privacy benefits.

Related Literature. Our study contributes to a large body of literature that explores the impact of public policies on venture investment and entrepreneurship (Howell 2017; Bloom, Van Reenen and Williams 2019). Specifically, we focus on the effects of data policies and regulation. A burgeoning stream of research has examined the implications of such policies in other domains. Goldfarb and Tucker (2011, 2012) examine the impact of the EU Privacy Directive on online advertising, demonstrating that the rollout of the Directive was associated with a decline in advertising effectiveness, thereby potentially reducing data monetization—a key component in the valuation of new data-driven ventures. Goldberg, Johnson and Shriver (2024) examine the impact of the GDPR on online web traffic, sales, and third-party tracking, demonstrating revenue declines due to the regulation. They further show that beliefs about local regulatory strictness may

influence firms' compliance activities.

Our study provides new and complementary insights from the perspective of investors. We show how the costs and uncertainty possibly brought about by privacy regulation (e.g., with respect to the adequacy of a venture's compliance, compliance costs, necessary product modifications, potential impact on data-driven revenues, and with respect to how larger platforms will pursue compliance) interface with investment strategies. Hence, any regulatory approach that aims to alleviate some of the costs identified in these related works has to be cognizant of the heterogeneous responses of investors.

Recent finance research connects data regulation to firm behavior and financial intermediation. Ramadorai, Uettwiller and Walther (2025) document heterogeneity in firms' data-extraction practices and changes in privacy disclosures around the GDPR and the California Consumer Privacy Act. He, Huang and Zhou (2023) show theoretically how customer-controlled data sharing changes lenders' screening advantages and credit-market competition, while Babina et al. (2025) find that open-banking policies stimulate venture investment in fintech startups. These studies highlight the importance of the rules governing access to and use of data for financial activity. We contribute by examining how privacy regulation affects the geographic allocation and organization of venture financing.

Closest to our paper, Jia, Jin and Wagman (2021) demonstrate that the rollout of the GDPR was associated with reductions in the number of EU venture deals, the size of those deals, and the overall amounts invested in EU ventures. Their focus is on the broader effects of data regulation on venture investment, whereas this paper emphasizes the interface between GDPR and investment strategies: we distinguish investor origins, follow the allocation of activity within a pre-existing investor cohort, and separate participation across financing roles. These analyses identify how the financing response varies with investor geography and responsibility for the investment.

Moreover, our study contributes to the literature on the organization of venture fi-

nancing and investor involvement. Lerner (1995) studies venture capitalists' oversight of private firms, and Gompers et al. (2020) document venture capitalists' roles in screening, structuring investments, and supporting portfolio companies. Amornsiripanitch, Gompers and Xuan (2019) show that lead-investor status and geographic proximity predict board representation, whereas Bernstein, Giroud and Townsend (2016) provides evidence that investor involvement improves portfolio-company innovation and exit outcomes. Syndication allows investors to combine information and resources in screening and supporting ventures (Lerner 1994; Brander, Amit and Antweiler 2002; Tian 2012), while assigning different responsibilities to lead and non-lead participants. More recent work examines how coordination costs and investors' incentives to share information shape syndicate organization (Kang, Li and Oh 2022; Luo 2025). We contribute by examining how foreign investment responds to regulatory exposure across these financing roles. The negative estimates for US lead and solo participation highlight primary investment responsibility as an important margin of adjustment. This distinction matters because participation in a financing round and leadership of that round represent different forms of financial intermediation.

Finally, our findings contribute to the literature on investors' local preferences and cross-border capital allocation.[9] Cumming and Dai (2010) documents local bias in venture capital investment, while related research connects geographic proximity to informational advantages in financial transactions (Kang and Kim 2008; Agarwal and Hauswald 2010). Bottazzi, Da Rin and Hellmann (2016) further shows that trust between countries predicts cross-border venture capital investment. We contribute by examining how a change in the regulatory environment affects the geographic allocation of entrepreneurial financing. Our evidence links regulatory exposure to a relative reduction in US participation in European ventures and to shorter investor–venture distances among completed investments. Following a pre-existing investor cohort also allows us

[9] For surveys, see Coeurdacier and Rey (2013) and Devigne et al. (2018).

to examine allocation across destinations conditional on investors' quarterly activity.

The remainder of the paper is organized as follows. Section 2 describes the data. Section 3 presents our empirical strategy. Section 4 examines the geography and composition of venture financing, and Section 5 follows individual investors and their financing roles. Section 6 considers the exclusion of UK ventures. Section 7 concludes.

2. Data and Measurement

2.1. Sample and units of observation

The primary source of data is PitchBook, which provides information on private-market financing rounds, the ventures receiving funding, and the investors participating in each round. We collect data on completed investment deals involving EU or US ventures from January 2014 through December 2019. We choose to stop the sample period in 2019 because the COVID shock starting in early 2020 has brought substantial changes to startup investment. For each financing round, the data contain venture information (name, headquarters location, business description, industry classification, and founding year) and funding information (the amount raised, financing date, funding stage, and participating investors). We link the transaction, company, and investor records using PitchBook's deal, company, and investor identifiers. These links allow us to distinguish the geography of the venture from that of its investors and to follow the same investor across financing rounds.

We restrict the sample to six PitchBook deal types: Accelerator/Incubator, Angel (individual), Seed Round, Early Stage VC, Later Stage VC, and PE Growth/Expansion. We date each transaction using its deal date and, when that date is unavailable, its announcement date. Appendix A documents these restrictions, record linkage, and the treatment of missing information.

Venture deals are often syndicated, making the identification of lead investors impor-

tant for distinguishing investors' roles. We identify the sole investor in a single-investor round as its lead. For rounds with multiple investors, we first use PitchBook's designated lead investor. When no lead is designated, we use the largest positive investor-specific dollar amount disclosed in PitchBook's deal–investor records; if no such amount identifies a lead, we use the earliest available investor founding year. We retain co-leads, including ties under either fallback rule. Lead status is thus partly inferred. These rules fail to identify a lead for 3,170 of the 135,712 linked rounds remaining after the institutional exclusions (2.34 percent), which we exclude.

We further exclude 7,270 rounds without an eligible EU or US investor and 226 US transactions with missing state information or locations outside the 50 states and the District of Columbia, leaving 125,046 deals: 51,956 involving European ventures and 73,090 involving US ventures. Throughout the analysis, a *state* denotes a US state (including the District of Columbia) or one of the 24 European countries in Appendix Table A.2. The UK is included in the European group throughout our 2014–2019 sample, as it remained an EU member during this period and was subject to the GDPR.

We aggregate financing rounds into an industry–state–quarter panel, retaining cells with no transactions as zeros for count outcomes. We also construct an investor–industry–state–quarter panel that counts unique investor–deal participations.[10] Industries are defined using PitchBook's Primary Industry Group classification, which includes broad categories such as Software, Retail, and Pharmaceuticals and Biotechnology.[11] A syndicated round contributes one deal to the aggregate count but one participation for each distinct investor. The investor panel retains 12,155 US and 6,327 European investors active before April 2016 and expands their observed industry–state relationships over all 24 quarters. Appendix A.1 details this construction.

[10] Among EU and US investors, we exclude universities, government entities, and sovereign wealth funds from the investor–industry–state–quarter panel. In the industry–state–quarter panel, deals involving these investors are retained if they meet the other sample criteria. Results remain robust if we exclude deals involving these investor types from the industry–state–quarter panel.

[11] Appendix Table A.6 lists all industry groups.

2.2. Exposure and outcome definitions

We measure industry exposure to the GDPR based on the role of personal data in ventures' products and services. General classifications of technology or data-intensive businesses need not capture this exposure: a business may use substantial computing capacity without depending heavily on personal information, while a retailer, recruitment service, or healthcare provider may routinely process it.

Following the description-based approach of Ewens, Nanda and Rhodes-Kropf (2018), we develop a dictionary of personal-data practices relevant to the GDPR (European Parliament and Council of the European Union 2016). We classify a venture as data-related if its PitchBook description matches at least one dictionary term. For each industry, we calculate the share of pre-enactment transactions, January 2014–March 2016, involving data-related ventures, pooling Europe and the US. A share of at least 50 percent identifies an exposed industry. This rule identifies 11 exposed industries among 40 with pre-enactment transactions, and the classification is fixed throughout the sample. It captures differences in industries' reliance on personal data, rather than whether individual firms are legally subject to the GDPR. Exposed industries include Retail, IT Services, and Media, while Computer Hardware, Semiconductors, and Energy Equipment fall below the cutoff.[12]

We distinguish five overlapping categories of deals. Defining US and EU as two separate unions, cross-union participation requires at least one investor from the venture's opposite union; cross-union leadership requires at least one lead from that union, including mixed co-leads. Same-union-led deals require confirmed same-union-only leads. EU-only deals have at least one EU investor and no identified US investor; US-only deals are defined symmetrically.[13]

[12]Company descriptions are available records, not historical snapshots at each financing date. Appendix A.3 documents the dictionary and industry classification.

[13]Third-country or unlocated investors may also participate in EU-only and US-only deals.

At the investor level, syndicate-lead, solo-investor, and syndicate non-lead participations partition total participation. We also distinguish initial from follow-on participation using the *New Investor* and *Follow-On Investor* statuses supplied by PitchBook for each investor–company relationship. Follow-on participation does not require the investor to have led an earlier round. Appendix A.2 provides detailed classification rules, including the treatment of missing statuses and the ordering of follow-on participation.

Counts are our primary outcomes because incomplete disclosure may bias comparisons of financing amounts. Disclosed round amounts are measured in millions of US dollars. At the investor level, because for most deals we only observe the total amount raised across all participating investors, we divide each round's total disclosed dollar amount equally across the all the investors participating in the round—including non-EU and non-US ones.[14] Cells containing undisclosed rounds are excluded from the investor-level amount regressions. These allocated amounts proxy for financing associated with participation, rather than observed investor-specific check sizes.

We measure investor–venture distance using headquarters coordinates. For investor i and venture j, the great-circle distance is

$$d_{ij} = 3963 \arccos\big[\sin(\text{lat}_i)\sin(\text{lat}_j) + \cos(\text{lat}_i)\cos(\text{lat}_j)\cos(\text{lon}_i - \text{lon}_j)\big], \tag{1}$$

where latitude and longitude are in radians and 3,963 is the Earth's radius in statute miles (Cumming and Dai 2010). We first average distances within each deal and role over EU and US investors, then give each complete deal mean equal weight within its industry–state–quarter cell. Distances describe completed investments, not the probability of investing at a particular distance. Appendix A.4 documents geocoding, missingness, dollar allocation, and macroeconomic controls.

[14] Investor-specific dollar amounts used to identify lead investors are too sparse for this purpose, covering only 15.54 percent of EU/US participations.

2.3. Deal and investor description

Table 1 links the structure of venture financing to the investors whose activity we follow. Panel A describes the full 2014–2019 financing sample. Cross-union investors participate in 15.87 percent of European rounds and 11.46 percent of US rounds, while cross-union leads appear in 11.27 and 5.77 percent, respectively. US rounds are more often syndicated (45.93 versus 35.00 percent). These distinctions motivate separate analyses of participation and leadership rather than treating all foreign involvement as the same outcome. The shares of data-related rounds are similar across the two markets.

Panel B describes the investor cohort before enactment. We use this earlier period to characterize investors' initial activity, geographic reach, and roles without incorporating their subsequent response to the GDPR. Mean participation counts exceed the median of one in both groups, indicating uneven activity across investors. The mean opposite-union share is 5.53 percent for US investors and 14.79 percent for European investors, reflecting geographic specialization. The role shares motivate the lead and solo comparisons, while the substantial angel and accelerator presence explains why the investor sample should not be read as a sample of VC firms alone.

Appendix Tables A.3 and A.4 provide additional statistics.

TABLE 1. Financing structure and the investor cohort

	Statistic	EU	US
Panel A: Financing sample, 2014Q1–2019Q4			
Financing rounds	Count	51,956	73,090
Industry–state–quarter cells	Count	23,616	50,184
Data-related rounds	Share (%)	36.18	36.09
Rounds with cross-union participation	Share (%)	15.87	11.46
Cross-union-led rounds	Share (%)	11.27	5.77
Syndicated rounds	Share (%)	35.00	45.93
Investors per round	Mean [median]	1.96 [1]	3.09 [1]
Panel B: Investor cohort and pre-enactment activity, 2014Q1–2016Q1			
Investors	Count	6,327	12,155
Participations per investor	Mean [median]	4.40 [1]	5.56 [1]
Venture states per investor	Mean [median]	1.52 [1]	2.20 [1]
Opposite-union participations	Mean share (%)	14.79	5.53
Data-exposed participations	Mean share (%)	27.84	24.12
Syndicate-lead participations	Mean share (%)	18.44	13.61
Solo participations	Mean share (%)	27.52	17.51
Angel / accelerator investors	Share (%)	36.12	43.78
Venture capital investors	Share (%)	27.22	27.41
Other investor types	Share (%)	36.67	28.80

Notes: Panel A describes the full financing sample by venture location; round shares use all rounds in each location as the denominator. Panel B describes investors active before April 2016 by investor location. Their activity is measured before enactment to separate initial investment patterns from subsequent adjustment. Mean participation shares give each investor equal weight; investor-type shares use PitchBook's primary classification for this cohort. Other types include PE/growth, corporate/CVC, and remaining types. Cross-union investment is between the EU and US. Means and bracketed medians summarize investment counts and geographic reach. All statistics precede estimation-specific sample restrictions. Appendix Tables A.3 and A.4 report the full distributions and quarterly panel statistics.

3. Empirical Strategy

We examine the effect of the GDPR using a triple-difference design that compares investment across venture locations, industry exposure, and policy periods. Comparing European and US ventures alone would combine the response to the regulation with other changes in the relative attractiveness of these markets. Comparing exposed and less-exposed industries within both regions accounts for geographic changes shared across

industries and exposure-related changes common to Europe and the US. The design asks whether investment in exposed European industries changed disproportionately after the GDPR.

We distinguish the enactment and transition period, from April 2016 through April 2018, from rollout, beginning in May 2018. Investors could adjust both as compliance requirements were clarified during the transition and once the regulation became enforceable. Monthly policy indicators are averaged within each quarter: enactment equals one between 2016Q2 and 2018Q1, while rollout equals one since 2018Q3 and 2018Q2 receives an enactment weight of one-third and a rollout weight of two-thirds. Both coefficients compare the relevant policy period with the period before April 2016.

3.1. Industry–state specification

Our preferred specification includes industry-by-state and state-by-quarter fixed effects. The former account for persistent differences in financing activity across industries and locations; the latter allow each European country and US state its own quarterly financing conditions. We estimate the following regression via Poisson Pseudo-Maximum Likelihood (PPML):

$$\begin{aligned}\mathbb{E}[\, y_{hst} \mid \cdot] = \exp\Big(&\alpha_{hs} + \alpha_{st} \\ &+ \gamma_4(Data_h \times GDPR_Enact_t) + \gamma_5(Data_h \times GDPR_Rollout_t) \\ &+ \beta_1(EU_s \times Data_h \times GDPR_Enact_t) \\ &+ \beta_2(EU_s \times Data_h \times GDPR_Rollout_t)\Big),\end{aligned} \tag{2}$$

where h denotes industry, s venture state, and t quarter. EU_s identifies European venture states, and $Data_h$ is the fixed pre-enactment exposure indicator. The exposure-by-policy-

period interactions capture changes common to exposed industries in Europe and the US. All other lower-order terms are absorbed by the fixed effects.

The coefficients β_1 and β_2 capture changes in the exposed-to-less-exposed investment ratio in Europe relative to the corresponding changes in the US. Our principal coefficient of interest is β_2, with $100[\exp(\beta_2) - 1]$ expressing the rollout differential in percentage terms. These estimates describe relative industry changes, rather than changes in total European financing.

Using the industry-state-quarter panel, Table 2 reports triple DID results for five outcomes: count of deals with cross-union participation, count of deals led by a cross-union investor, count of deals lead by a same-union investor, count of deals with only EU investors, and count of deals with only US investors. For cross-union outcomes, the comparison includes US investment in European ventures and EU investment in US ventures.

For comparison, the first five columns of Table 2 report a specification with additive industry, state, and quarter effects, which are less saturated than our preferred specification with industry-state and state-quarter fixed effects in Columns 6-10.[15] We retain industry-by-state and state-by-quarter effects for the remaining industry-level analyses and apply the same three-way comparison to disclosed amounts and continuous-exposure checks.

Our PPML regressions accommodate zero investment cells without taking logarithms of counts. Standard errors are two-way clustered by industry and venture state, allowing dependence across states within an industry and across industries within a state.

3.2. Investor specification

The investor panel follows investors active before April 2016 within their observed industry–state relationships. We estimate the policy and exposure comparisons sepa-

[15]The additive specification includes all estimable lower-order interactions and controls for unemployment, GDP-per-capita growth, CPI inflation, and the long-term interest rate. State-by-quarter effects absorb these controls.

rately for US and European investors. The outcome y_{ihst} counts distinct investor–deal participations, where i indexes investors. Our preferred specification is

$$
\begin{aligned}
\mathbb{E}[\, y_{ihst} \mid \cdot] = \exp\Bigg(& \alpha_{ihs} + \alpha_{st} + \lambda_{it} \\
& + \gamma_4 (Data_h \times GDPR_Enact_t) + \gamma_5 (Data_h \times GDPR_Rollout_t) \\
& + \beta_1 (EU_s \times Data_h \times GDPR_Enact_t) \\
& + \beta_2 (EU_s \times Data_h \times GDPR_Rollout_t) \Bigg).
\end{aligned} \tag{3}
$$

Investor-by-industry-by-state effects α_{ihs} account for an investor's persistent presence in a particular industry and venture state. State-by-quarter effects capture local financing conditions, while investor-by-quarter effects λ_{it} absorb changes in an investor's overall quarterly activity. The estimates therefore describe the relative allocation of investment across observed venture industries and states, conditional on investor activity in that quarter. Each investor table also reports the specification without investor-by-quarter effects. Standard errors are again clustered by venture industry and state.

3.3. Identification and dynamics

Identification requires that, absent the GDPR, the exposed-to-less-exposed investment ratio would have evolved similarly in European and US venture states, conditional on the included fixed effects. Contemporaneous changes specific to exposed European industries could violate this assumption. The estimates also allow for effects of the regulation on US ventures or less-exposed industries: they measure a relative exposure effect, rather than an effect against a wholly unaffected comparison group.

To examine dynamics of the differential effects of GDPR in exposed industries while limiting the imprecision associated with estimating separate quarterly coefficients, we complement the estimates over the enactment and rollout periods with event-study

specifications using annual time interactions and 2015 as the reference year. These retain quarterly observations and the corresponding fixed effects while imposing a common coefficient across quarters within each calendar year. Appendix D reports the pre-period diagnostics and their limitations.[16]

4. The GDPR and the Geography of Venture Financing

4.1. Financing across exposed industries

Table 2 summarizes the estimated effects of the GDPR on foreign venture financing across industries. The key coefficients of interest are the triple interactions. These capture how the change in financing from before GDPR to the enactment and rollout periods differs between exposed and less-exposed industries in Europe, compared with the corresponding difference in the US.

In the additive specification, the rollout estimates imply relative reductions of approximately 12.5 percent in rounds with cross-union participation, 28.3 percent in cross-union-led rounds, and 18.2 percent in US-only rounds (columns 1, 2, and 5). All three estimates are significant at the 1 percent level. The corresponding estimates for same-union-led and EU-only rounds are negative but statistically insignificant (columns 3 and 4). The positive EU$\times$Rollout coefficients for cross-union participation, cross-union leadership, and US-only financing indicate increases in financing-round counts for less-exposed European industries relative to the corresponding US outcomes.

The estimates from our preferred specification are similar. Including industry-by-state and state-by-quarter fixed effects, we find that GDPR rollout led to relative reductions of 10.95 percent in rounds with cross-union participation and 19.20 percent in US-only rounds (columns 6 and 10). The estimated reduction in cross-union-led rounds is larger,

[16]The displayed dynamics are drawn from count and distance outcomes whose static rollout triple interaction in the preferred specification is significant at the 10 percent level, together with the distance outcomes. Static tables retain both significant and insignificant central outcomes.

Table 2. European financing growth and industry exposure

	Industry, state, and quarter FE					Industry × state, state × quarter FE				
	(1) Cross-union particip.	(2) Cross-union led	(3) Same-union led	(4) EU-only	(5) US-only	(6) Cross-union particip.	(7) Cross-union led	(8) Same-union led	(9) EU-only	(10) US-only
EU × *Data*	0.299***	0.342***	0.346***	0.538***	0.225	–	–	–	–	–
	(0.102)	(0.107)	(0.093)	(0.140)	(0.143)					
EU × *Enactment*	0.163***	0.146***	0.117*	0.028	0.256***	–	–	–	–	–
	(0.051)	(0.040)	(0.063)	(0.065)	(0.049)					
EU × *Rollout*	0.221***	0.228***	0.180**	0.043	0.378***	–	–	–	–	–
	(0.058)	(0.050)	(0.072)	(0.079)	(0.066)					
Data × *Enactment*	-0.213***	-0.159**	-0.203***	-0.097	-0.201***	-0.225***	-0.169**	-0.208***	-0.129	-0.205***
	(0.047)	(0.070)	(0.054)	(0.124)	(0.061)	(0.055)	(0.076)	(0.055)	(0.141)	(0.059)
Data × *Rollout*	-0.302***	-0.175	-0.235***	-0.212	-0.228***	-0.330***	-0.199	-0.242***	-0.266	-0.234***
	(0.085)	(0.131)	(0.074)	(0.142)	(0.083)	(0.090)	(0.138)	(0.079)	(0.169)	(0.084)
EU × *Data* × *Enactment*	-0.043	-0.149*	0.025	-0.075	-0.071	-0.042	-0.157*	0.032	-0.041	-0.080
	(0.081)	(0.084)	(0.054)	(0.102)	(0.061)	(0.083)	(0.091)	(0.059)	(0.120)	(0.082)
EU × *Data* × *Rollout*	-0.133***	-0.332***	-0.033	-0.053	-0.201***	-0.116***	-0.325***	-0.021	0.007	-0.213***
	(0.042)	(0.122)	(0.054)	(0.106)	(0.035)	(0.034)	(0.121)	(0.056)	(0.142)	(0.034)
Industry FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
State FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Quarter FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Industry × state FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
State × quarter FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Observations before pruning	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000
Observations in estimation	66,576	66,576	72,000	66,456	72,000	21,805	17,960	48,001	23,217	39,437

Notes: Outcomes count financing rounds and may overlap. Cross-union leadership requires at least one opposite-union lead; same-union leadership requires confirmed same-union-only leads. EU-only and US-only exclude investors from the opposite union but allow third-country or unlocated investors. All estimates are from Poisson regressions. *EU* identifies European venture states; *Data* is industry exposure fixed before April 2016. Enactment covers April 2016–April 2018 and Rollout begins in May 2018; their weights in 2018Q2 are 1/3 and 2/3. Standard errors are two-way clustered by venture industry and state. ***, **, and * denote significance at the 1, 5, and 10 percent levels. A dash denotes an absorbed coefficient or control, or an excluded fixed effect. Positive-semidefinite covariance correction applies in columns 1, 2, 3, 4, 5, 6, 10.

at 27.76 percent (column 7). All three estimates are significant at the 1 percent level and economically substantial in magnitude. The corresponding estimates for same-union-led and EU-only rounds are close to zero and statistically insignificant, providing little evidence of a similar relative contraction in financing involving investors from the venture's own union.

The larger leadership point estimate for the decrease in cross-union financing in exposed industries in the EU is consistent with regulatory frictions weighing particularly on investment relationships that require greater investor involvement.[17] Leading an investment typically involves responsibility for evaluating the venture, negotiating terms, and coordinating financing. A relative reduction in foreign-led rounds may therefore affect access to these services alongside access to capital. The aggregate estimates do not establish whether foreign investors shifted from leading rounds to participating alongside other investors; the investor-level analysis provides complementary evidence by distinguishing these roles.

The estimates of triple interaction terms concerning the GDPR enactment provide more limited evidence of adjustment. Cross-union leadership exhibits a negative enactment triple interaction significant at the 10 percent level in both specifications, whereas the corresponding enactment interactions for the other four outcomes are statistically insignificant. Annual estimates likewise show negative leadership and US-only coefficients in 2017, before enforcement, followed by negative point estimates in 2018 and 2019 (Appendix Figure D.1). Thus, the relative declines in cross-union participation and US-only financing are statistically significant only during rollout, while cross-union leadership also shows some evidence of adjustment during the transition to enforcement.

Overall, the estimates indicate a statistically and economically significant relative contraction in foreign financing for exposed European industries following GDPR rollout.

[17] The deal categories overlap, so their coefficients cannot be added. Comparisons of coefficient magnitudes are descriptive and do not establish statistically significant differences across outcomes.

4.2. Leadership within syndicates

The decline in foreign leadership could reflect fewer solo investments or a change in who leads syndicated rounds. To better understand how the GDPR reshaped cross-border investment flows between EU and US, in Table 3, we examine GDPR-induced changes in syndication behavior. Syndication plays a crucial role in mitigating investment risks and reducing screening and monitoring costs (Lerner 1994). The regulatory uncertainty introduced by the GDPR may increase the overall tendency of investors to syndicate deals as a risk management strategy.

We find that neither the total number of syndicated rounds nor cross-union participated syndicated rounds changed significantly during rollout. However, column 4 shows a large (37.52 percent) and statistically significant decline in syndicated rounds led by a cross-union investor. The leadership pattern therefore extends to rounds in which investors share financing with others.

Moreover, given that the uncertainties associated with the GDPR are likely more pronounced for non-European investors, we examine whether US investors increase their reliance on partnering with European investors as non-lead investors. Local partners bring essential expertise in navigating complex regulatory environments and managing GDPR compliance, making them invaluable collaborators for foreign investors aiming to sustain their investment activities in the European market. Nevertheless, column 3 of Table 3 shows that cross-union participation under same-union leadership has a positive but insignificant coefficient. Thus, the decline in foreign leadership does not come with evidence of an offsetting increase alongside local leads. Appendix Figure D.1 reports event-study evidence in support of these findings.

TABLE 3. Cross-union participation and leadership in syndicates

	(1) All	(2) Cross-union particip.	(3) Same-union led	(4) Cross-union led
Data × *Enactment*	-0.230***	-0.251***	-0.301***	-0.179***
	(0.050)	(0.043)	(0.083)	(0.055)
Data × *Rollout*	-0.308***	-0.366***	-0.430***	-0.197
	(0.079)	(0.081)	(0.092)	(0.144)
EU × *Data* × *Enactment*	0.132	-0.024	0.118	-0.297
	(0.094)	(0.115)	(0.130)	(0.204)
EU × *Data* × *Rollout*	0.053	-0.087	0.176	-0.470***
	(0.092)	(0.081)	(0.125)	(0.155)
Industry × state FE	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes
Macroeconomic controls	–	–	–	–
Observations before pruning	72,000	72,000	72,000	72,000
Observations in estimation	35,972	17,259	12,088	11,171

Notes: Outcomes count syndicated rounds. The last two columns restrict cross-union participation by the location of leads; mixed co-leads enter cross-union leadership. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Dashes mark absorbed terms or excluded fixed effects. Positive-semidefinite covariance correction applies in columns 3, 4.

4.3. Distance and the composition of completed investments

The decline in foreign leadership raises a related question: do the investors in completed rounds become more local? Proximity can reduce screening and monitoring costs, while unfamiliar legal and institutional environments can make distant investments more costly. Cumming and Dai (2010) document home bias in US venture investment, particularly among technology investors. Familiarity can also influence investment choice (Huberman 2001; Franke et al. 2006), including through lower information-acquisition costs. If compliance uncertainty reinforces the value of proximity, the average distance between investors and exposed European ventures could shrink.

Table 4 estimates Equation 2 using the average investor–venture distance in miles as outcome. The rollout estimates imply relative reductions of 18.45 percent for all investors and 25.17 percent for leads, both significant at the 1 percent level. The non-lead estimate

TABLE 4. Investor–venture distance

	(1) All investors	(2) Leads	(3) Non-leads
Data × *Enactment*	-0.041	-0.058	-0.042
	(0.047)	(0.040)	(0.063)
Data × *Rollout*	-0.054	-0.019	-0.075
	(0.050)	(0.047)	(0.047)
EU × *Data* × *Enactment*	-0.090	-0.094	-0.070
	(0.126)	(0.131)	(0.093)
EU × *Data* × *Rollout*	-0.204***	-0.290***	0.009
	(0.071)	(0.076)	(0.066)
Industry × state FE	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes
Macroeconomic controls	–	–	–
Observations before pruning	20,427	20,426	11,386
Observations in estimation	19,977	19,967	10,741

Notes: Outcomes are cell-average distances in miles for EU and US investors. Each complete deal mean receives equal weight; cells without usable distances are excluded. Distances describe completed investments. All estimates are from Poisson regressions. *EU* identifies European venture states; *Data* is industry exposure fixed before April 2016. Enactment covers April 2016–April 2018 and Rollout begins in May 2018; their weights in 2018Q2 are 1/3 and 2/3. Standard errors are two-way clustered by venture industry and state. ***, **, and * denote significance at the 1, 5, and 10 percent levels. A dash denotes an absorbed coefficient or control, or an excluded fixed effect.

is close to zero. Annual lead-distance coefficients are negative from 2016 onward, with confidence intervals excluding zero in 2018 and 2019 (Appendix Figure D.2). These patterns indicate that the decline in foreign-led financing is accompanied by a shift toward geographically closer lead investors among completed rounds.

4.4. Financing across stages

We next examine which stages of venture financing are affected by the foreign investment pullback. Table 5 reports estimates from our industry–state–quarter specification separately for seed and startup, early-stage VC, later-stage VC, and PE/growth rounds.

TABLE 5. Financing rounds by stage

	Seed and Startup (1)	Early-stage VC (2)	Later-stage VC (3)	PE growth/ expansion (4)
Panel A: Cross-union participation				
EU × *Data* × *Enactment*	-0.026	-0.096*	0.005	0.386
	(0.152)	(0.056)	(0.174)	(0.471)
EU × *Data* × *Rollout*	-0.071***	-0.109	-0.187	-0.061
	(0.026)	(0.139)	(0.158)	(0.290)
Observations before pruning	72,000	72,000	72,000	72,000
Observations in estimation	14,417	10,840	10,333	1,582
Panel B: Cross-union leadership				
EU × *Data* × *Enactment*	-0.210	0.085	-0.186	-0.120
	(0.182)	(0.243)	(0.351)	(1.008)
EU × *Data* × *Rollout*	-0.272***	-0.171	-0.497	-0.509
	(0.065)	(0.235)	(0.308)	(0.549)
Observations before pruning	72,000	72,000	72,000	72,000
Observations in estimation	12,198	7,043	6,950	1,021
Panel C: US-only financing				
EU × *Data* × *Enactment*	-0.043	-0.177	-0.114	0.678
	(0.148)	(0.164)	(0.352)	(0.697)
EU × *Data* × *Rollout*	-0.233***	-0.065	-0.449**	0.751
	(0.075)	(0.144)	(0.210)	(0.627)
Observations before pruning	72,000	72,000	72,000	72,000
Observations in estimation	34,209	21,688	19,541	7,705
Industry × state FE	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes
Macroeconomic controls	–	–	–	–

Notes: Outcomes count distinct rounds in industry–state–quarter cells in the full aggregate sample. Seed and Startup pools Accelerator/Incubator, Angel, and Seed; early-stage VC, later-stage VC, and PE Growth/Expansion are separate. Accelerator/Incubator/Angel and venture capital investors account for 54.00 and 36.09 percent of EU and US investor–deal participations in Seed and Startup rounds, together representing the majority (90.09 percent). Each round belongs to one stage but may enter multiple financing outcomes. Zero-investment cells are retained before estimation-specific pruning. Estimates are from Poisson regressions with industry-by-state and state-by-quarter effects and exposure-by-policy-period interactions. Policy definitions and industry/state clustering follow Section 3. Macroeconomic controls are absorbed. Stars denote significance at the 1, 5, and 10 percent levels. Clustered covariance matrices use positive-semidefinite correction where required.

Seed and startup financing comprises accelerator/incubator, angel, and seed rounds, regardless of the types of investors participating.[18]

[18]This category contains 71,173 rounds and 122,796 investor–deal participations by EU and US investors. Angels and accelerators account for 54.00 percent of these participations and VC investors for 36.09 percent, or 90.09 percent combined.

For seed and startup rounds, the rollout estimates imply relative declines of 6.85 percent in cross-union participation, 23.81 percent in cross-union leadership, and 20.78 percent in US-only financing, all significant at the 1 percent level. The pullback therefore reaches the earliest financing stages, including rounds in which foreign investors take a lead role. US-only financing also declines by 36.17 percent in later-stage VC rounds, significant at the 5 percent level. This later-stage decline is particularly relevant to the concerns about scale-up financing of European ventures, given the role of US investors in financing their expansion. Overall, however, the pullback reaches both the earliest financing stages and later-stage rounds that can support further growth.

5. Investor Participation and the Organization of Financing

The preceding results describe changes in financing received by ventures. We now follow individual investors active before April 2016 to examine how the pullback appears within the existing investor population. Foreign financing can decline when investors reduce their overall activity or redirect it across destinations. Its consequences also depend on the roles investors undertake: participation alongside another lead differs from assuming primary responsibility for evaluating a venture and organizing its financing. We examine these two dimensions by studying investors' allocation of activity and their participation across financing roles.

5.1. Participation and allocation across destinations

Table 6 examines whether exposed European industries receive relatively less participation from established US investors after accounting for changes in those investors' overall activity. Investor-by-quarter fixed effects absorb changes in each investor's total quarterly participation in the observed sample, while investor-by-industry-by-state and state-by-quarter effects account for persistent investment specialization and local

financing conditions. The specification therefore distinguishes changes in the allocation of investors' activity from a general expansion or contraction in their participation. With these effects, US participation in exposed European industries declines by 11.77 percent relative to the comparison groups during rollout (-0.125, significant at the 10 percent level). The negative estimate indicates that the foreign financing pullback includes a relative shift away from exposed European industries within established US investors' activity. The estimate without investor-by-quarter effects is similar, implying a 12.55 percent decline (-0.134, significant at the 5 percent level)..[19]

For European investors, the rollout coefficient is 0.020 with or without investor-by-quarter effects and is statistically insignificant. We therefore find evidence of reduced participation by US investors in exposed European industries, but no statistically detectable change in the corresponding European-investor outcome.

Allocated amounts associated with US participation also decline, by 46.63 percent with investor-by-quarter effects (-0.628, significant at the 1 percent level; Appendix C). These estimates indicate that the participation decline is accompanied by a reduction in the deal amounts allocated to US investors under our measurement procedure. They do not measure changes in investors' actual capital contributions, which are not observed.[20]

We also examine investors observed investing after enactment or rollout (Appendix Table B.2). With investor-by-quarter effects, the rollout coefficient is -0.124 among investors active from 2016Q2 onward and -0.125 among those active from 2018Q2 onward. The former is significant at the 10 percent level; the latter narrowly misses that threshold ($p = 0.1054$). These negative point estimates are consistent with continuing

[19]The paired specifications retain different samples: adding investor-by-quarter effects reduces the retained observations from 1,865,828 to 1,057,574 and the number of investors from 12,155 to 7,883. The coefficient comparison therefore reflects changes in both the specification and the estimation sample and does not provide a decomposition of the decline into overall activity and allocation.

[20]Appendix Table C.6 reports allocated-amount estimates by investor type. The negative VC and private-equity/growth estimates become smaller and insignificant with investor-by-quarter effects, while the corporate/CVC estimate is positive and significant. Appendix C discusses amount disclosure, allocation, and estimation-sample differences.

TABLE 6. Investor participation by investor location

	US		EU	
	(1)	(2)	(3)	(4)
Data × *Enactment*	-0.250***	-0.238***	-0.260***	-0.259***
	(0.045)	(0.044)	(0.055)	(0.060)
Data × *Rollout*	-0.328***	-0.323***	-0.307***	-0.304***
	(0.081)	(0.083)	(0.070)	(0.096)
EU × *Data* × *Enactment*	0.003	0.017	0.122	0.119**
	(0.056)	(0.055)	(0.079)	(0.054)
EU × *Data* × *Rollout*	-0.134**	-0.125*	0.020	0.020
	(0.068)	(0.074)	(0.053)	(0.055)
Investor × industry × state FE	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–
Observations before pruning	1,870,536	1,870,536	772,584	772,584
Observations in estimation	1,865,828	1,057,574	766,363	378,432
Investors in estimation	12,155	7,883	6,326	3,995

Notes: The samples consist of US and European investors active before April 2016. Each outcome counts distinct investor–deal participations. All estimates are from Poisson regressions. *EU* identifies European venture states; *Data* is industry exposure fixed before April 2016. Enactment covers April 2016–April 2018 and Rollout begins in May 2018; their weights in 2018Q2 are 1/3 and 2/3. Standard errors are two-way clustered by venture industry and state. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Paired columns omit and include investor-by-quarter effects, respectively. Both retain investor-by-industry-by-state and state-by-quarter effects; macroeconomic controls are absorbed. Retained observations and investor counts are outcome-specific; the paired models are not estimated on a common retained sample. Positive-semidefinite covariance correction applies in column 4.

investors reallocating activity away from exposed European industries, although the evidence is less precise for the rollout-active sample. Because these restrictions condition on subsequent investment in either union, they provide descriptive checks rather than a separate identification strategy.[21]

[21]The investment-relationship results do not establish whether the adjustment occurs primarily through new or existing investor–venture relationships (Appendix Table B.4). With investor-by-quarter effects, the initial-participation rollout coefficient is −0.092 and insignificant. Follow-on, second, and third-or-later participation coefficients are also negative but insignificant.

5.2. Leadership and participation alongside another lead

The distinction between financing roles helps connect these participation results to the organization of foreign investment. Leads and solo investors take primary responsibility for an investment, while non-leads participate alongside another investor who leads the round. A decline in foreign leadership therefore concerns a different form of investor involvement from a decline in participation alone. Table 7 separates US investor participations into syndicate leads, solo investors, and syndicate non-leads.

In these separate regressions, investor-by-quarter effects absorb each investor's total quarterly participation in the role being examined. The estimates therefore ask whether US investors direct relatively less of their lead, solo, or non-lead activity toward exposed European industries, conditional on how often they undertake that role overall. With these effects, syndicate-lead participation exhibits a relative decline of 32.05 percent during rollout (-0.386), while solo-investor participation declines by 29.42 percent (-0.348). Both estimates are significant at the 5 percent level. The syndicate non-lead estimate is positive and statistically insignificant (0.083). The specifications without investor-by-quarter effects show the same qualitative pattern.[22]

The clearest evidence of reduced US participation thus occurs in roles carrying primary investment responsibility. This gives the aggregate leadership findings an investor-level counterpart: among investors active before the GDPR, participation in exposed European ventures declines when US investors lead a syndicate or invest alone. The non-lead estimate provides no statistically significant evidence of increased participation alongside another lead.

The event-studies in Figure 1 provide the clearest evidence of persistent declines in US lead and solo participation. Syndicate-lead coefficients are negative in 2017–2019, with confidence intervals excluding zero in 2018 and 2019. Solo coefficients are negative,

[22]The regressions retain different samples across roles and do not directly test differences between the role-specific coefficients.

Table 7. US investor participation by role

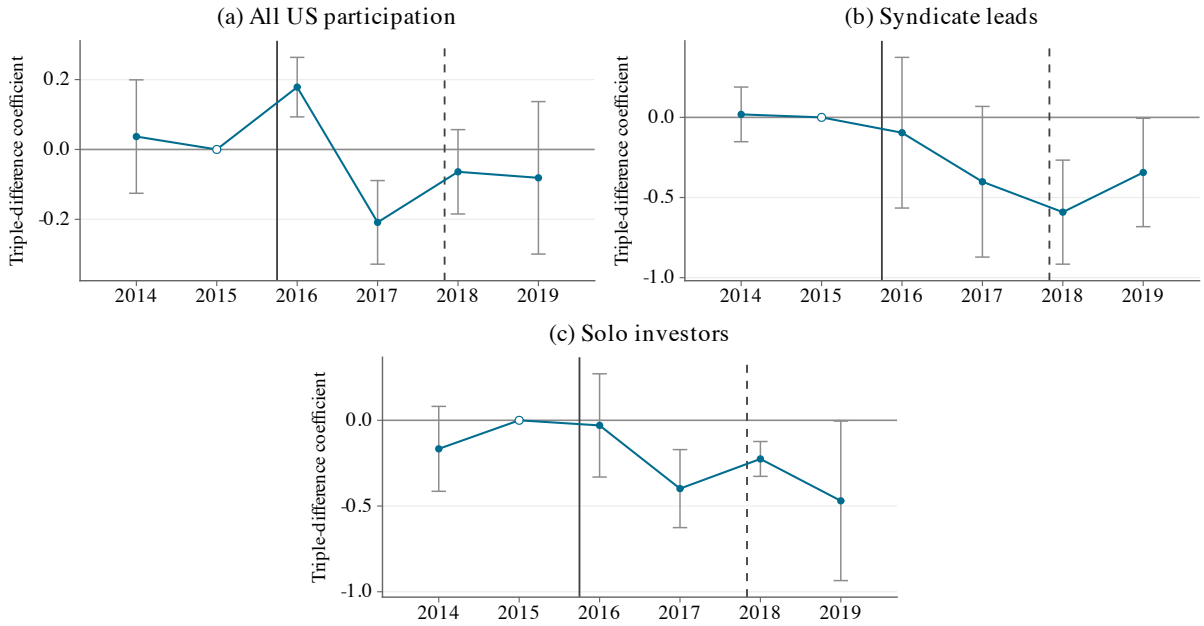
	Syndicate lead		Solo		Syndicate non-lead	
	(1)	(2)	(3)	(4)	(5)	(6)
Data × *Enactment*	-0.260***	-0.266***	-0.226***	-0.173**	-0.248***	-0.253***
	(0.050)	(0.055)	(0.068)	(0.082)	(0.049)	(0.047)
Data × *Rollout*	-0.350***	-0.376***	-0.190***	-0.127	-0.361***	-0.382***
	(0.069)	(0.085)	(0.073)	(0.096)	(0.093)	(0.090)
EU × *Data* × *Enactment*	-0.279	-0.282	-0.064	-0.096	0.153*	0.207**
	(0.180)	(0.176)	(0.097)	(0.106)	(0.084)	(0.101)
EU × *Data* × *Rollout*	-0.338***	-0.386**	-0.286***	-0.348**	0.023	0.083
	(0.113)	(0.152)	(0.076)	(0.167)	(0.120)	(0.136)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–
Observations before pruning	1,870,536	1,870,536	1,870,536	1,870,536	1,870,536	1,870,536
Observations in estimation	466,574	204,694	473,554	193,310	1,262,510	627,741
Investors in estimation	4,734	2,621	4,732	2,391	10,488	6,461

Notes: The sample consists of US investors active before April 2016. Lead roles are recorded or inferred as described in Appendix A. No direct cross-role coefficient test is imposed by these separate regressions. All estimates are from Poisson regressions. *EU* identifies European venture states; *Data* is industry exposure fixed before April 2016. Enactment covers April 2016–April 2018 and Rollout begins in May 2018; their weights in 2018Q2 are 1/3 and 2/3. Standard errors are two-way clustered by venture industry and state. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Paired columns omit and include investor-by-quarter effects, respectively. Both retain investor-by-industry-by-state and state-by-quarter effects; macroeconomic controls are absorbed. Retained observations and investor counts are outcome-specific; the paired models are not estimated on a common retained sample. Positive-semidefinite covariance correction applies in column 3.

with confidence intervals excluding zero, in all three years. For total US participation, the coefficient is negative in 2017 but closer to zero and imprecise in 2018–2019. The annual evidence thus shows a more persistent reduction in participation involving primary investment responsibility.

These findings connect the geographic allocation of foreign financing to the responsibilities investors assume. The declines in lead and solo participation are consistent with the argument that regulatory uncertainty can weigh on foreign investors' willingness to evaluate and oversee exposed European ventures. The estimates do not directly identify changes in screening or monitoring costs. They do show that the foreign financing pullback includes reduced participation in roles involving primary investment responsi-

FIGURE 1. Annual dynamics of US investor participation



Notes: Panels show all US participations, syndicate-lead participations, and solo participations. All models include investor-by-industry-by-state, state-by-quarter, and investor-by-quarter effects. The models retain quarterly observations and impose common event coefficients within each calendar year, with 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout within the calendar years. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with standard errors clustered by industry and state and corrected for a non-positive-semidefinite covariance matrix where required. Table D.1 reports the annual pre-period tests.

bility, a dimension of financial intermediation that aggregate participation counts alone would not reveal.

6. Robustness and Limits of Interpretation

We examine sensitivity to exposure measurement and the inclusion of UK ventures.

Continuous exposure. Replacing the binary indicator with standardized continuous exposure yields relative rollout declines of 14.45 percent in cross-union participation, 24.02 percent in cross-union leadership, and 20.18 percent in US-only financing, all significant at the 1 percent level (Appendix Table B.1). The negative static pattern therefore extends beyond the binary cutoff. However, the participation and leadership

pre-period tests reject ($p = 0.0273$ and 0.0279), qualifying their causal interpretation. The US-only test does not reject ($p = 0.8379$; Appendix Figure D.4). These diagnostics are limited by the available annual pre-period comparison between 2014 and 2015.

The role of the UK. We retain UK ventures in the main sample because the UK remained an EU member throughout 2014–2019 and was subject to the GDPR from May 2018. Brexit became effective only on January 31, 2020, after our sample ended. Nevertheless, EU law, and hence the GDPR, continued to apply in the UK through 31 December 2020, until it was replaced by the UK General Data Protection Regulation (UK GDPR).[23] Including the UK therefore follows the regulation's geographic coverage during the sample period. The UK is also a major European venture-capital hub, and excluding it substantially reduces the financing activity represented in the sample. The June 2016 Brexit referendum preceded GDPR rollout by nearly two years, whereas our main financing results are clearest during rollout. Moreover, state-by-quarter fixed effects include UK-by-quarter effects, absorbing Brexit-related shocks common across UK industries.

Nevertheless, as an additional precaution, we exclude UK ventures to assess whether the results depend on this market or on UK-specific changes affecting exposed and less-exposed industries differently. Such differential changes would not be absorbed by UK-by-quarter effects. We retain UK investors and the full-sample industry exposure classification in this exercise. We find that the exclusion preserves the relative decline in US-only financing: the rollout estimate implies a 21.18 percent reduction, significant at the 1 percent level (Appendix Table B.5). With investor-by-quarter effects, the syndicate-lead coefficient is nearly unchanged, from -0.386 to -0.392, although less precisely estimated and no longer statistically significant. Total and solo participation attenuate to -0.081 and -0.140 and are also insignificant. The aggregate decline thus extends

[23] See https://www.gov.uk/data-protection for details.

beyond the UK's distinctive financing and political setting, while the leadership estimate remains similar in magnitude.

Together, these checks leave the evidence strongest for the relative decline in US-only financing. Pre-period differences under continuous exposure and the weaker investor-level evidence outside the UK qualify the broader interpretation. The available annual pre-period comparison, between 2014 and 2015, also limits the assessment of parallel trends.

7. Conclusion

This paper examines the impact of the GDPR on venture investment flows between the EU and US. Using data on venture investments from 2014 to 2019, this paper documents a relative contraction in US venture financing for data-exposed European industries around the implementation of the GDPR.

The introduction of the GDPR imposed additional regulatory burdens on technology ventures, raising uncertainty about investment returns. In response, we find that investors adjusted their strategies by prioritizing geographically closer ventures and, when outside the EU, reduced their participation in roles involving primary investment responsibility. While increases in the supply of capital from EU investors or from syndicates involving both US and EU investors represent mechanisms that may partially mitigate the adverse effects, we find no evidence indicating significant shifts in these margins.

Our findings highlight the important, unintended consequences of the GDPR on transatlantic (cross-union) investments, particularly by US-based investors, suggesting that privacy regulation has implications for the organization of cross-border investment. Changes in foreign investors' willingness to assume primary investment responsibility may matter beyond the financing attached to their participation. Lead investors can contribute expertise, coordinate financing, and help ventures access networks and in-

ternational markets. Changes in these relationships could therefore affect the support available to entrepreneurs alongside their access to capital.

These considerations are particularly relevant for Europe, where US investors complement a comparatively less developed domestic venture-capital ecosystem. Evaluating entrepreneurial financing requires attention to who provides it and the responsibilities those investors undertake. Financing volumes alone may not fully capture changes in the expertise and international connections available to ventures.

For policymakers, the findings motivate considering access to foreign capital and investment leadership alongside the intended privacy benefits of regulation. The broader challenge is to pursue privacy protection while sustaining the investment relationships that support entrepreneurial activity. Understanding how regulatory design and implementation influence these relationships can contribute to a fuller assessment of privacy policy and its consequences for innovation.

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Appendix A. Data Construction and Measurement

A.1. Sample construction

We retain completed Accelerator/Incubator, Angel (individual), Seed Round, Early Stage VC, Later Stage VC, and PE Growth/Expansion deals with a usable deal date or, if absent, an announcement date during 2014–2019. Deal, company, and investor identifiers link the records. Venture headquarters must be in the US or a country in Table A.2. Among EU and US investors, University, Government, and Sovereign Wealth Fund records are excluded. Table A.1 reconciles the deal restrictions and constructed panels.

TABLE A.1. Sample construction

Restriction or panel	Rounds / observations
Completed eligible financing rounds dated 2014–2019	280,448
EU or US venture headquarters	172,858
At least one linked participation after institutional exclusions	135,712
At least one identified lead after all fallback rules	132,542
At least one eligible EU or US investor	125,272
Valid venture state for the quarterly panels	125,046
European venture rounds	51,956
US venture rounds	73,090
Unique EU/US investor–deal participations	284,569
Constructed industry–state–quarter cells	73,800
Industry cells with classified pre-enactment exposure	72,000
Constructed pre-active investor–industry–state–quarter cells	2,643,120

Notes: The first six rows apply restrictions sequentially to financing rounds. Lead identification excludes 3,170 of 135,712 linked rounds (2.34 percent). The final location restriction excludes 226 US rounds. Investor participations count distinct investor–deal pairs across eligible EU and US investors, before restricting to the pre-enactment cohort. Constructed panel counts include zero-investment cells and precede outcome-specific missingness and fixed-effect pruning.

PitchBook Primary Industry Group defines industries. Two otherwise unclassified ventures receive explicit assignments: Kava Biosciences to Pharmaceuticals and Biotechnology and Pro-Gest to Consumer Non-Durables. The state restriction excludes missing US states and locations outside the 50 states and the District of Columbia. European countries are treated as states. The industry panel expands 41 industries across 75 states

TABLE A.2. List of EU states included in the sample

Austria	Belgium	Croatia	Czech Republic
Denmark	Estonia	Finland	France
Germany	Greece	Hungary	Republic of Ireland
Italy	Latvia	Luxembourg	Netherlands
Poland	Portugal	Romania	Slovakia
Slovenia	Spain	Sweden	United Kingdom

and 24 quarters, including zeros. Other Healthcare remains in the constructed descriptive panel but has no exposure measure and is omitted from exposure regressions.

The deal–investor links contain 329,982 participation records on the final deals. The eligible EU/US subset contains 287,313 records; consolidating 2,744 repeated investor–deal records leaves 284,569 distinct participations by 37,446 investors. Multiple funds or partner records may contribute to the reconstructed deal-level syndicate count but do not create repeated participations for the same investor in the investor panel. Lead status at the consolidated investor–deal level indicates whether any retained record identifies that investor as a lead.

For the investor panel, we identify every investor–industry–state combination observed at any point during 2014–2019 and expand it over all 24 quarters. Count outcomes are zero in quarters without investment. A relationship first observed after enactment therefore includes its earlier zero quarters; the set of destinations is not restricted to pre-enactment investments.

We deliberately omit combinations never observed during the sample to avoid constructing all-zero groups. Both investor specifications include a separate fixed effect for each investor–industry–state combination. In Poisson regression, the fixed effect for a group with zero investment in every quarter tends to negative infinity, making its fitted counts tend to zero regardless of the slope coefficients. Such groups provide no information about those coefficients and are removed from estimation. Expanding all possible

destinations would therefore leave the slope estimates unchanged under these specifications while greatly increasing the size of the constructed panel. This result depends on the combination fixed effects, not on the additional investor-by-quarter effects.

We retain EU and US investors active before April 2016 and relationships with classified exposure, leaving 12,155 US and 6,327 European investors. Their 110,130 observed relationships produce 2,643,120 cells. Missing values are handled separately for each outcome, including unknown relationship status, ambiguous investment order, and undisclosed amounts. Poisson estimation applies further sample restrictions according to the outcome and fixed effects. Tables A.3 and A.4 describe the full constructed data; regression tables report retained estimation samples.

Table A.3. Summary statistics for venture financing

	EU ventures					US ventures				
	Mean	SD	Median	95th pct.	N	Mean	SD	Median	95th pct.	N
Panel A: Industry–state–quarter level										
# of deals	2.20	10.39	0	9	23,616	1.46	12.01	0	5	50,184
# of deals with a cross-union investor	0.35	2.54	0	1	23,616	0.17	1.95	0	1	50,184
# of cross-union-led deals	0.25	1.83	0	1	23,616	0.08	0.93	0	0	50,184
# of same-union-led deals	1.91	8.53	0	8	23,616	1.33	10.71	0	5	50,184
# of EU-only deals	1.85	8.08	0	8	23,616	0.05	0.54	0	0	50,184
# of US-only deals	0.18	1.39	0	1	23,616	1.29	10.14	0	5	50,184
# of syndicated deals	0.77	4.10	0	3	23,616	0.67	6.66	0	2	50,184
# of cross-union syndicated deals	0.19	1.43	0	1	23,616	0.13	1.59	0	0	50,184
# of cross-union-led syndicated deals	0.09	0.70	0	0	23,616	0.05	0.56	0	0	50,184
# of same-union-led cross-union syndicated deals	0.09	0.72	0	0	23,616	0.07	0.98	0	0	50,184
$MM in deals with a cross-union investor	3.22	37.75	0	3.05	23,616	4.29	71.68	0	0	50,184
$MM in cross-union-led deals	2	30.59	0	0.15	23,616	1.71	46.41	0	0	50,184
$MM in same-union-led deals	3.82	26.52	0	13.96	23,616	9.54	117.54	0	20.78	50,184
$MM in EU-only deals	3.09	21.25	0	11.87	23,616	0.25	9.23	0	0	50,184
$MM in US-only deals	0.59	10.86	0	0	23,616	8.17	100.65	0	19.10	50,184
Investor–venture miles: all investors	581.32	950.24	201.75	2,447.09	7,895	812.90	817.28	666.41	2,368.28	12,539
Investor–venture miles: lead investors	581.38	1,012.90	171.78	2,673.54	7,917	814.90	898.81	605.87	2,489.03	12,516
Investor–venture miles: non-lead investors	608.31	976.22	201.01	2,744.45	4,392	975.38	802.70	883.66	2,415.32	6,996
Data-exposed industry	27.50%	44.65%	0%	100%	23,040	27.50%	44.65%	0%	100%	48,960
Pre-enactment data-related share	30.21%	24.05%	22.49%	71.65%	23,040	30.21%	24.05%	22.49%	71.65%	48,960
Panel B: State–quarter level										
Unemployment rate	8.34%	4.72%	6.87%	19.30%	576	4.49%	1.27%	4.30%	6.70%	1,224
GDP per capita growth (YoY %)	4.07%	4%	3.26%	9.61%	576	2.99%	2.37%	3.14%	5.93%	1,224
CPI inflation (YoY %)	1%	1.12%	0.88%	2.97%	576	1.55%	0.93%	1.65%	3.05%	1,224
Interest rate	1.64%	1.60%	1.05%	4.42%	576	2.32%	0.39%	2.27%	2.93%	1,224
Panel C: Deal level										
Data-related deal	36.18%	48.05%	0%	100%	51,956	36.09%	48.03%	0%	100%	73,090
Syndication	35%	47.70%	0%	100%	51,956	45.93%	49.83%	0%	100%	73,090
Cross-union syndication	8.69%	28.17%	0%	100%	51,956	8.92%	28.51%	0%	100%	73,090
# of investors	1.96	1.95	1	6	51,956	3.09	3.81	1	11	73,090
# of cross-union investors	0.22	0.69	0	1	51,956	0.16	0.54	0	1	73,090
# of same-union investors	1.53	1.36	1	4	51,956	2.53	2.98	1	9	73,090
Amount per deal ($MM)	4.97	31.85	0.69	16.96	30,015	12.64	89.08	2	49.40	49,487

Notes: This table describes the industry–state–quarter panel, 2014Q1–2019Q4, separately for European and US ventures. States are US states (including the District of Columbia) and European countries. Panel A covers all 73,800 constructed cells, Panel B unique state–quarters, and Panel C the underlying deals. Each row uses its nonmissing values. Cross-union denotes at least one opposite-union investor; same-union participation is not required. Cross-union-led deals include mixed co-leads; same-union-led deals have confirmed same-union-only leads. EU-only (US-only) deals have at least one European (US) investor and none from the other union. Syndicated deals have at least two participation records. Amounts sum disclosed round amounts in millions of US dollars ($MM). Distances average complete deal-level role means over EU and US investors; unavailable distances are missing. Description keywords identify data-related deals. Data-exposed industries have a pre-enactment (2014Q1–2016Q1) data-related transaction share of at least 50 percent; the continuous measure is this share. Both exposure measures are missing for Other Healthcare. Investor counts include third-country participation records. Amount per deal excludes undisclosed rounds. Indicators and shares are percentages; macroeconomic variables are percentages as published. For percentage variables, SD is measured in percentage points.

Table A.4. Summary statistics for investors

	US investors					EU investors				
	Mean	SD	Median	95th pct.	N	Mean	SD	Median	95th pct.	N
Panel A: Investor level, 2014Q1–2016Q1										
# of participations	5.56	16.64	1	22	12,155	4.40	11.26	1	17	6,327
# of venture industries	2.38	2.68	1	8	12,155	2.34	2.63	1	8	6,327
# of venture states	2.20	2.61	1	7	12,155	1.52	1.50	1	4	6,327
Opposite-union participation share	5.53%	20.23%	0%	50%	12,155	14.79%	32.04%	0%	100%	6,327
Data-exposed participation share	24.12%	35.31%	0%	100%	12,155	27.84%	37.69%	0%	100%	6,327
Syndicate-lead share	13.61%	27.95%	0%	100%	12,155	18.44%	32.31%	0%	100%	6,327
Solo-investor share	17.51%	33.86%	0%	100%	12,155	27.52%	39.57%	0%	100%	6,327
Pre-VC share	40.01%	43.07%	25%	100%	12,155	39.64%	44.13%	14.29%	100%	6,327
Later-stage VC share	27.18%	38.93%	0%	100%	12,155	25.96%	38.61%	0%	100%	6,327
Angel / accelerator	43.78%	49.61%	0%	100%	12,155	36.12%	48.04%	0%	100%	6,327
Venture capital	27.41%	44.61%	0%	100%	12,155	27.22%	44.51%	0%	100%	6,327
Private equity / growth	7.93%	27.02%	0%	100%	12,155	9.02%	28.66%	0%	100%	6,327
Corporate / CVC	11.66%	32.09%	0%	100%	12,155	17.81%	38.26%	0%	100%	6,327
Panel B: Investor level, 2014Q1–2019Q4										
# of industry–state relationships	6.41	12.04	3	23	12,155	5.09	7.58	2	17	6,327
Any European-venture relationship	20.24%	40.18%	0%	100%	12,155	92.70%	26.02%	100%	100%	6,327
Any data-exposed European-venture relationship	7.37%	26.13%	0%	100%	12,155	54.70%	49.78%	100%	100%	6,327
Active from 2018Q2 onward	46.63%	49.89%	0%	100%	12,155	47.38%	49.94%	0%	100%	6,327
Panel C: Investor–industry–state–quarter level										
# of participations	0.09	0.40	0	1	1,870,536	0.09	0.39	0	1	772,584
Any participation	7.51%	26.35%	0%	100%	1,870,536	7.98%	27.10%	0%	100%	772,584
# of syndicate-lead participations	0.02	0.14	0	0	1,870,536	0.02	0.16	0	0	772,584
# of solo-investor participations	0.02	0.21	0	0	1,870,536	0.03	0.25	0	0	772,584
# of syndicate non-lead participations	0.05	0.26	0	0	1,870,536	0.04	0.21	0	0	772,584
# of initial participations	0.07	0.34	0	1	1,870,308	0.08	0.34	0	1	772,475
# of follow-on participations	0.02	0.16	0	0	1,870,308	0.02	0.15	0	0	772,475
# of pre-VC participations	0.04	0.28	0	0	1,870,536	0.04	0.26	0	0	772,584
# of early-stage VC participations	0.03	0.18	0	0	1,870,536	0.03	0.19	0	0	772,584
# of later-stage VC participations	0.02	0.17	0	0	1,870,536	0.02	0.16	0	0	772,584
European venture state	7.18%	25.82%	0%	100%	1,870,536	81.63%	38.73%	100%	100%	772,584
Data-exposed industry	26.95%	44.37%	0%	100%	1,870,536	30.69%	46.12%	0%	100%	772,584
European venture state × data-exposed industry	1.95%	13.82%	0%	0%	1,870,536	26.39%	44.07%	0%	100%	772,584

Notes: The panel includes 12,155 US and 6,327 European investors with at least one participation before April 2016. Each observed investor–industry–state relationship is expanded over all 24 quarters from 2014Q1 to 2019Q4, independently of estimation-specific sample restrictions. Panel A describes pre-enactment participations, 2014Q1–2016Q1; each share weights participations equally within investor and the reported statistics weight investors equally. Opposite-union denotes ventures in the other union; data-exposed denotes industries with pre-enactment data-related shares of at least 50 percent. Syndicate leads invest with others and lead; solo investors finance a round alone. Pre-VC pools Accelerator/Incubator, Angel (individual), and Seed Round; later-stage VC pools Later Stage VC and PE Growth/Expansion. Types follow PitchBook's primary classification; Angel / accelerator includes angels, angel groups, and accelerators/incubators, and Corporate / CVC combines corporate/strategic and corporate venture capital investors. Other types are not displayed. Panel B describes full-period relationships and activity; active from 2018Q2 means any participation from that quarter onward. Panel C covers investor–industry–state–quarter cells. Lead, solo, and non-lead participations sum to total participations. Initial/follow-on use the source New Investor and Follow-On Investor statuses; cells with unknown status are excluded from these two rows. Each row uses its nonmissing values. Indicators and shares are percentages. For percentage variables, SD is measured in percentage points.

A.2. Outcome and role definitions

We distinguish five categories of deals. Cross-union participation requires at least one identified investor from the other union; cross-union leadership requires at least one lead investor from the other union. Same-union-led deals require all identified leads to belong to the venture's union; a lead whose location is unknown or outside the EU and US prevents this classification. For example, a French venture financed by a German lead and a US non-lead is both same-union-led and cross-union-participated. A deal with both a French and a US lead is cross-union-led. We additionally identify EU-only deals as those with at least one EU investor and no identified US investor, and define US-only deals symmetrically. These last two categories may include third-country investors or investors with unknown locations. The five categories are not mutually exclusive.

Syndication uses the number of participation records after the institutional exclusions. A round with a single remaining record is assigned a lead. In multi-investor rounds, recorded leads take precedence. If none is designated, the largest positive value of PitchBook's *InvestorInvestmentAmount* identifies a lead; if no positive amount is available, the earliest known founding year among the eligible EU/US investor records is used. Ties create co-leads. This field contains disclosed investor-specific dollar amounts when available, but is too sparse to serve as the monetary outcome: coverage among EU/US investor–deal participations is 15.54 percent. Unresolved lead status excludes the round.

At the investor level, syndicate leads, solo investors, and syndicate non-leads partition each eligible investor's participations. A round led entirely by third-country or unlocated investors can still contribute EU/US non-lead participations. Initial and follow-on status comes from *InvestorStatus*: *New Investor* and *Follow-On Investor*. Repeated source statuses are consolidated within investor–deal pairs; unknown or conflicting statuses cause the containing cell to be excluded from the initial/follow-on decomposition.

Second and third-or-later participations are constructed from completed eligible fi-

nancings for the same investor–company pair, including pre-2014 transactions. Deal dates take precedence over announcement dates. We count strictly prior dated transactions, rather than ranking only transactions within the estimation window. A follow-on with no observed predecessor, a pair with undated eligible history, or same-day transactions spanning an ordinal boundary prevents reliable assignment. Same-day blocks beginning at the third participation remain safely third-or-later. Cells with unresolved follow-on order are excluded from the ordinal decomposition. Initial/follow-on status is not inferred from lead status.

A.3. Exposure construction

The exposure measure links the use of personal data described in PitchBook company records to the activities addressed by the GDPR. The steps below document the dictionary, industry shares, and comparison with the cloud-computing classification.

Our approach is motivated by Ewens, Nanda and Rhodes-Kropf (2018), who use company descriptions to distinguish industry segments that benefited from the reduction in experimentation costs following the introduction of cloud computing. Their classification identifies businesses likely to benefit from lower costs of online infrastructure, storage, and computing, using the prevalence of terms such as online, e-commerce, hosting, and web. We follow their approach of using firms' activities to characterize industry exposure, but develop a separate dictionary for the personal-data practices relevant to the GDPR.

We construct the dictionary by reading the provisions of Regulation (EU) 2016/679 governing personal information and its processing (European Parliament and Council of the European Union 2016). The terms cover customer and user identifiers, direct marketing, profiling and tracking, health and location information, online transactions, financial and employment records, and services that process data for other businesses. For example, customer data and CRM indicate the use of identifiable customer records,

while patient portals, payroll, and recruitment describe activities that routinely involve personal information. These terms translate the regulation's concepts into the language used in company descriptions. Appendix Table A.5 reports the dictionary, the associated provisions, and the empirical rationale for each group. The keywords measure likely exposure rather than a firm's legal obligations or compliance.

We apply the dictionary to lowercase PitchBook descriptions and classify a venture as data-related if at least one term matches. Word stems capture common variants. Terms match as substrings, except that CRM requires a word boundary on both sides and creator, ride, VoIP, and loan require a word boundary at the start. Missing descriptions are treated as empty strings and do not match. Each transaction inherits its venture's classification; industry names alone do not determine it.

We next fix industry exposure using the 41,743 transactions observed from January 2014 through March 2016, before GDPR enactment. For each PitchBook Primary Industry Group, we calculate the share of these transactions involving data-related ventures, pooling the EU and US samples. An industry is classified as data-exposed when this share is at least 50 percent. The classification applies to European and US venture states and remains fixed, so post-enactment changes in the composition of financed ventures do not alter industry exposure. Because the measure is based on transactions, a venture raising several rounds contributes once for each round.[24]

The rule identifies 11 exposed industries among the 40 with pre-enactment transactions. Other Healthcare, the remaining industry in the 41-industry panel, has no pre-enactment transaction and cannot be assigned an exposure value. We impose no minimum transaction count; Other Information Technology consequently enters the exposed group on the basis of one pre-enactment deal. Appendix Table A.6 reports all industry shares and their denominators, including separate EU and US shares. We also stan-

[24]The pre-enactment restriction applies to the transactions used to calculate industry shares. Business descriptions are drawn from the available PitchBook company records; they are not separately archived descriptions observed at each financing date.

dardize the 40 nonmissing industry shares, giving industries equal weight, to assess sensitivity to the binary cutoff. The classification is intended to measure relative exposure: industries below the cutoff may contain data-related firms, and the cutoff does not separate firms that are legally subject to the GDPR from those that are not.

Software illustrates the distinction between a technology industry and personal-data exposure. Of its 15,979 pre-enactment transactions, 41.6 percent involve descriptions matching the dictionary, placing it below the 50 percent cutoff (Table A.6). Software remains in the estimation sample; the binary indicator does not imply that its firms are unaffected by the GDPR. The continuous specification retains its measured share and uses variation across all 40 classified industries, avoiding a discrete assignment at the cutoff. The rollout estimates with industry-by-state and state-by-quarter effects remain negative and significant for cross-union participation, cross-union leadership, and US-only financing (Appendix Table B.1). The check supports the main financing pattern under a continuous measure. It addresses sensitivity to the cutoff, not the accuracy of the dictionary or the legal exposure of software firms. The continuous cross-union pre-period tests reject, as discussed in Section 6 and Appendix D.

TABLE A.5. GDPR Provisions and the Keyword Dictionary

Provision	Empirical cue	Derived keywords
Article 4(1)	Personal data and identifiers	customer data, user data, crm
Articles 6(1)(a), 7, and 21	Consent, withdrawal of consent, and direct marketing	marketer, advertis, campaign, newsletter, loyalty, monetiz
Articles 4(1), 4(4), and 22, with Recital 30	Online identifiers, profiling, tracking, and automated individual decisions	cookie, targeting, personaliz, profil, audience, first-party, influencer, creator
Articles 4(15) and 9	Health data and special categories of personal data	telehealth, telemedicine, health record, appointment, care coordination, patient engagement, home care, caregiver, clinic management, health service, patient portal
Article 4(1)	Location data and services that routinely process location or movement information	booking, reservation, passenger, ride, travel, delivery, commuter
Article 4(1), with Recital 30	Subscriber records, online identifiers, and electronic-communications services	telecom, broadband, messaging, voip, connectivity, subscriber base, network service
Article 6	Customer-facing transactions for which processing requires a lawful basis	e-commerce, ecommerce, online shop, online store, online retail, online market, online service, online order, online food, merchant, shopper, marketplace, checkout, point-of-sale, storefront, subscriber, membership, streaming, viewers, publish, fashion, apparel, footwear, retailer
Article 6	Financial transactions and identity-linked customer records	loan, lender, payment
Article 88	Personal data in employment and recruitment contexts	recruit, talent, payroll, staffing
Articles 4(8) and 28	Firms that process personal data on behalf of business customers	it service, managed service, managed it, cloud service, software development, system integrat, systems integration, it consult, it outsourc, it infrastructure, it solution, digital transformation, help desk, network management, data center, cybersecurity service, tech support, consulting, outsourc, professional service, technology service, it support, software solution, application development

Notes: The entries link provisions of Regulation (EU) 2016/679 to terms in business descriptions. They are empirical proxies rather than statutory definitions. Terms are matched jointly to lowercase PitchBook descriptions; stems such as advertis, personaliz, and profil capture variants. A match to any term identifies a data-related venture. Industry names do not override the description-based classification.

Table A.6. Pre-Enactment Data-Related Exposure by Industry

Industry	All ventures	US ventures	EU ventures
Other Information Technology	100.0% (1)	100.0% (1)	N/A
Retail	78.7% (760)	81.8% (351)	76.0% (409)
Apparel and Accessories	64.6% (664)	66.2% (429)	61.7% (235)
Other Financial Services	60.2% (635)	65.9% (422)	48.8% (213)
Communications and Networking	60.0% (408)	58.8% (204)	61.3% (204)
Transportation	57.5% (449)	53.2% (216)	61.4% (233)
Restaurants, Hotels and Leisure	56.8% (650)	57.1% (289)	56.5% (361)
IT Services	56.2% (557)	53.6% (291)	59.0% (266)
Commercial Banks	55.6% (27)	62.5% (16)	45.5% (11)
Commercial Services	52.9% (4,373)	55.1% (2,362)	50.3% (2,011)
Media	52.9% (1,833)	52.4% (1,036)	53.5% (797)
Healthcare Services	42.1% (818)	44.3% (571)	36.8% (247)
Software	41.6% (15,979)	41.3% (10,124)	42.3% (5,855)
Services (Non-Financial)	37.5% (1,423)	36.6% (809)	38.6% (614)
Healthcare Technology Systems	34.5% (1,433)	37.1% (1,083)	26.3% (350)
Other Consumer Products and Services	33.3% (3)	100.0% (1)	0.0% (2)
Commercial Transportation	30.5% (128)	36.5% (74)	22.2% (54)
Textiles	29.6% (27)	33.3% (15)	25.0% (12)
Insurance	27.0% (115)	31.9% (72)	18.6% (43)
Capital Markets/Institutions	23.4% (137)	18.8% (85)	30.8% (52)
Other Business Products and Services	21.6% (74)	26.8% (41)	15.2% (33)
Consumer Durables	20.6% (1,563)	20.8% (965)	20.2% (598)
Metals, Minerals and Mining	20.0% (30)	23.1% (13)	17.6% (17)
Consumer Non-Durables	19.5% (1,287)	18.0% (822)	22.2% (465)
Utilities	18.3% (60)	14.3% (42)	27.8% (18)
Semiconductors	17.8% (191)	20.5% (112)	13.9% (79)
Pharmaceuticals and Biotechnology	15.5% (2,043)	17.2% (1,380)	11.9% (663)
Computer Hardware	14.8% (1,202)	14.1% (615)	15.5% (587)
Energy Services	12.9% (194)	15.2% (105)	10.1% (89)
Containers and Packaging	11.5% (61)	9.5% (21)	12.5% (40)
Healthcare Devices and Supplies	11.4% (2,076)	11.7% (1,354)	10.7% (722)
Commercial Products	10.2% (1,438)	11.3% (699)	9.2% (739)
Agriculture	4.6% (196)	4.4% (114)	4.9% (82)
Energy Equipment	4.5% (396)	3.5% (202)	5.7% (194)
Forestry	3.7% (27)	0.0% (6)	4.8% (21)
Exploration, Production and Refining	3.6% (111)	4.2% (48)	3.2% (63)
Chemicals and Gases	3.1% (255)	1.9% (158)	5.2% (97)
Construction (Non-Wood)	0.0% (19)	0.0% (9)	0.0% (10)
Other Energy	0.0% (21)	0.0% (12)	0.0% (9)
Other Materials	0.0% (79)	0.0% (41)	0.0% (38)
Other Healthcare	N/A	N/A	N/A

Notes: Entries give the share of transactions from January 2014 through March 2016 involving ventures whose business descriptions match at least one dictionary term. Transaction counts appear in parentheses. The pooled EU and US share determines exposure; separate EU and US shares are descriptive. The dotted line marks the 50 percent cutoff. N/A indicates no observed transaction. No minimum transaction count is imposed. Other Healthcare has no pre-enactment observation and is excluded from exposure-based regressions.

A.4. Distance and other variables

We measure geographic proximity using headquarters locations from PitchBook's company and investor records. European locations are matched to representative city coordinates in GeoNames. US locations are matched by five-digit ZIP code to the 2019 Census ZCTA Gazetteer; missing, invalid, conflicting, or unmatched ZIP codes fall back to a city match within the reported state (U.S. Census Bureau 2019; GeoNames 2026). City names and aliases are matched within country and, where available, administrative subdivision, with postal information helping to resolve ambiguity. Unresolved locations remain missing. The resulting coordinates represent cities or ZIP areas, rather than street addresses, and reflect the available headquarters records rather than historical office locations.

Within each deal, we calculate separate average distances for all EU and US investors, leads, and non-leads using Equation 1. Each eligible investor receives equal weight, including co-leads; third-country investors are excluded. A role-specific deal mean is missing if that group is absent or any required coordinate is unavailable. We average complete deal means within industry–state–quarter cells with equal weight across deals. Larger syndicates therefore do not receive greater cell weight. Valid zero distances are retained; cells without usable distances remain missing. Frozen city-match overrides resolve reviewed location names; unmatched or ambiguous locations without a resolved match remain missing.

Disclosed round amounts enter industry-level dollar outcomes; an undisclosed amount does not remove a deal from counts. Investor-level dollar outcomes divide each disclosed round by its reconstructed participation-record count, including third-country and unlocated records. Duplicate records for the same investor are consolidated before cell aggregation. Cells containing an undisclosed round are excluded from investor monetary outcomes, so these samples differ from the count samples. This allocation is distinct from the disclosed investor-specific amounts used in lead inference.

Finally, we gather macroeconomic indicators for the countries and US states in which ventures are headquartered. These include the unemployment rate, consumer price index (CPI), long-term interest rate, and GDP per capita. European series come from Eurostat. For the US, unemployment and price data come from the Bureau of Labor Statistics, state GDP and population data from the Bureau of Economic Analysis, and the 10-year Treasury yield from Federal Reserve Economic Data. We match these series to the venture's location and transaction month, using national or euro-area measures where more geographically disaggregated series are unavailable. The main specifications use unemployment, year-on-year CPI inflation, year-on-year GDP-per-capita growth, and the long-term interest rate, averaged over the quarter for the quarterly analysis.

Appendix B. Additional Count Results and Robustness

The tables below report the continuous-exposure and additional investor results. The continuous-exposure table retains both industry specifications reported in Table 2. Each investor table compares the specification without investor-by-quarter effects with the specification including them. Both include investor-by-industry-by-state and state-by-quarter effects, but may retain different samples.

Table B.1. Venture financing and continuous industry exposure

	Industry, state, and quarter FE					Industry × state, state × quarter FE				
	(1) Cross-union particip.	(2) Cross-union led	(3) Same-union led	(4) EU-only	(5) US-only	(6) Cross-union particip.	(7) Cross-union led	(8) Same-union led	(9) EU-only	(10) US-only
EU × *Data*	0.249**	0.298***	0.050	0.047	0.302***	–	–	–	–	–
	(0.123)	(0.108)	(0.113)	(0.129)	(0.113)					
EU × *Enactment*	0.201***	0.179***	0.091*	0.003	0.281***	–	–	–	–	–
	(0.040)	(0.035)	(0.049)	(0.066)	(0.037)					
EU × *Rollout*	0.242***	0.239***	0.152**	0.017	0.404***	–	–	–	–	–
	(0.050)	(0.066)	(0.068)	(0.084)	(0.056)					
Data × *Enactment*	-0.050	-0.046	-0.140***	-0.034	-0.152***	-0.067	-0.056	-0.157***	-0.046	-0.158***
	(0.038)	(0.036)	(0.028)	(0.040)	(0.032)	(0.044)	(0.037)	(0.030)	(0.054)	(0.032)
Data × *Rollout*	-0.090	-0.064	-0.137***	-0.083	-0.150***	-0.118	-0.075	-0.160***	-0.116	-0.163***
	(0.076)	(0.088)	(0.045)	(0.082)	(0.049)	(0.074)	(0.090)	(0.049)	(0.112)	(0.050)
EU × *Data* × *Enactment*	-0.155**	-0.209***	0.086*	-0.013	-0.093*	-0.148**	-0.221***	0.109**	0.001	-0.118*
	(0.064)	(0.055)	(0.052)	(0.063)	(0.056)	(0.073)	(0.058)	(0.049)	(0.069)	(0.065)
EU × *Data* × *Rollout*	-0.168***	-0.261***	0.032	-0.017	-0.190***	-0.156***	-0.275***	0.060	0.014	-0.225***
	(0.054)	(0.087)	(0.048)	(0.078)	(0.046)	(0.058)	(0.089)	(0.048)	(0.102)	(0.043)
Industry FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
State FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Quarter FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Industry × state FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
State × quarter FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Observations before pruning	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000
Observations in estimation	66,576	66,576	72,000	66,456	72,000	21,805	17,960	48,001	23,217	39,437

Notes: Outcomes count financing rounds and may overlap. Cross-union leadership requires at least one opposite-union lead; same-union leadership requires confirmed same-union-only leads. EU-only and US-only exclude investors from the opposite union but allow third-country or unlocated investors. *Data* is the pre-enactment data-related share standardized across the 40 classified industries; coefficients represent a one-standard-deviation contrast. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Dashes mark absorbed terms or excluded fixed effects. Positive-semidefinite covariance correction applies in columns 1, 2, 3, 4, 5, 7, 10.

Table B.2. Participation among continuing US investors

	All pre-active		Active from 2016Q2		Active from 2018Q2	
	(1)	(2)	(3)	(4)	(5)	(6)
Data × *Enactment*	-0.250***	-0.238***	-0.238***	-0.238***	-0.235***	-0.233***
	(0.045)	(0.044)	(0.047)	(0.044)	(0.048)	(0.047)
Data × *Rollout*	-0.328***	-0.323***	-0.317***	-0.323***	-0.306***	-0.321***
	(0.081)	(0.083)	(0.081)	(0.082)	(0.083)	(0.084)
EU × *Data* × *Enactment*	0.003	0.017	0.003	0.018	0.001	0.018
	(0.056)	(0.055)	(0.051)	(0.055)	(0.053)	(0.059)
EU × *Data* × *Rollout*	-0.134**	-0.125*	-0.134**	-0.124*	-0.128**	-0.125
	(0.068)	(0.074)	(0.067)	(0.074)	(0.054)	(0.077)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–
Observations before pruning	1,870,536	1,870,536	1,723,608	1,723,608	1,564,176	1,564,176
Observations in estimation	1,865,828	1,057,574	1,718,992	1,050,895	1,559,708	1,015,414
Investors in estimation	12,155	7,883	7,516	7,131	5,668	5,518

Notes: Subsequent activity can occur in either venture union. These restrictions condition on post-treatment behavior and are descriptive. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Paired columns omit and include investor-by-quarter effects. Other fixed effects are shown, and macroeconomic controls are absorbed. Observation and investor counts are outcome-specific; pairs do not use a common retained sample.

Table B.3. US investor participation by investor type

	Angel / accel.		VC		PE / growth		Corporate / CVC	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Data × *Enactment*	-0.268***	-0.215***	-0.245***	-0.260***	-0.316***	-0.380***	-0.190***	-0.138**
	(0.084)	(0.059)	(0.053)	(0.059)	(0.093)	(0.124)	(0.049)	(0.059)
Data × *Rollout*	-0.271***	-0.213**	-0.359***	-0.394***	-0.469***	-0.587***	-0.230**	-0.143
	(0.105)	(0.083)	(0.088)	(0.087)	(0.133)	(0.134)	(0.108)	(0.129)
EU × *Data* × *Enactment*	-0.057	-0.002	0.040	0.056	-0.350***	-0.397***	0.043	0.227
	(0.067)	(0.108)	(0.091)	(0.072)	(0.125)	(0.070)	(0.118)	(0.201)
EU × *Data* × *Rollout*	-0.347***	-0.294***	0.018	-0.027	-0.298	-0.432	-0.253	-0.028
	(0.083)	(0.101)	(0.098)	(0.091)	(0.199)	(0.312)	(0.294)	(0.347)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–	–	–
Observations before pruning	650,472	650,472	831,960	831,960	122,160	122,160	139,896	139,896
Observations in estimation	644,959	305,824	825,822	562,801	115,566	56,226	133,088	63,272
Investors in estimation	5,322	3,148	3,331	2,715	963	652	1,416	691

Notes: Investor types follow the primary PitchBook classification. The sample consists of pre-enactment US investors in the indicated type. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Paired columns omit and include investor-by-quarter effects. Other fixed effects are shown, and macroeconomic controls are absorbed. Observation and investor counts are outcome-specific; pairs do not use a common retained sample. Positive-semidefinite covariance correction applies in columns 4, 5, 6.

Table B.4. Initial and follow-on US investor participation

	All		Initial		Follow-on		Second		Third or later	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Data × *Enactment*	-0.250***	-0.238***	-0.240***	-0.221***	-0.280***	-0.323***	-0.321***	-0.364***	-0.188*	-0.325***
	(0.045)	(0.044)	(0.052)	(0.047)	(0.075)	(0.078)	(0.081)	(0.083)	(0.102)	(0.102)
Data × *Rollout*	-0.328***	-0.323***	-0.310***	-0.304***	-0.363***	-0.424***	-0.352***	-0.383***	-0.398***	-0.556***
	(0.081)	(0.083)	(0.088)	(0.090)	(0.075)	(0.080)	(0.062)	(0.086)	(0.119)	(0.092)
EU × *Data* × *Enactment*	0.003	0.017	0.003	0.043	-0.122	-0.284	-0.125	-0.279	0.139	-0.040
	(0.056)	(0.055)	(0.018)	(0.046)	(0.340)	(0.438)	(0.295)	(0.379)	(0.344)	(0.492)
EU × *Data* × *Rollout*	-0.134**	-0.125*	-0.117**	-0.092	-0.336	-0.448	-0.394	-0.497	-0.137	-0.386
	(0.068)	(0.074)	(0.057)	(0.088)	(0.230)	(0.360)	(0.247)	(0.384)	(0.301)	(0.466)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–	–	–	–	–
Observations before pruning	1,870,536	1,870,536	1,870,308	1,870,308	1,870,308	1,870,308	1,869,160	1,869,160	1,869,160	1,869,160
Observations in estimation	1,865,828	1,057,574	1,766,682	933,546	413,193	193,432	363,762	146,816	128,917	50,043
Investors in estimation	12,155	7,883	11,801	7,580	4,478	2,322	4,151	2,089	1,815	859

Notes: Initial and follow-on status comes from PitchBook. Second and third-or-later participation is constructed from dated eligible histories, including pre-2014 deals. Unknown status and ambiguous order are excluded from the relevant cells. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Paired columns omit and include investor-by-quarter effects. Other fixed effects are shown, and macroeconomic controls are absorbed. Observation and investor counts are outcome-specific; pairs do not use a common retained sample. Positive-semidefinite covariance correction applies in columns 3, 4, 5, 6, 8, 10.

Table B.5. Financing and participation excluding UK ventures

Panel A: US investor participations

	Enactment				Rollout			
	Full		No UK		Full		No UK	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All participation	0.003	0.017	-0.072	-0.040	-0.134**	-0.125*	-0.099	-0.081
	(0.056)	(0.055)	(0.090)	(0.095)	(0.068)	(0.074)	(0.121)	(0.129)
Syndicate lead	-0.279	-0.282	-0.496**	-0.510***	-0.338***	-0.386**	-0.426**	-0.392
	(0.180)	(0.176)	(0.212)	(0.134)	(0.113)	(0.152)	(0.194)	(0.265)
Solo investor	-0.064	-0.096	-0.170	-0.134	-0.286***	-0.348**	-0.222**	-0.140
	(0.097)	(0.106)	(0.135)	(0.215)	(0.076)	(0.167)	(0.094)	(0.168)
Syndicate non-lead	0.153*	0.207**	0.163	0.257	0.023	0.083	0.107	0.137
	(0.084)	(0.101)	(0.179)	(0.226)	(0.120)	(0.136)	(0.190)	(0.254)
Active from 2016Q2	0.003	0.018	-0.062	-0.038	-0.134**	-0.124*	-0.089	-0.079
	(0.051)	(0.055)	(0.082)	(0.095)	(0.067)	(0.074)	(0.117)	(0.129)
Active from 2018Q2	0.001	0.018	-0.071	-0.036	-0.128**	-0.125	-0.107	-0.078
	(0.053)	(0.059)	(0.100)	(0.105)	(0.054)	(0.077)	(0.118)	(0.132)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes

Panel B: US-only financing rounds

	Enactment		Rollout	
	Full	No UK	Full	No UK
US-only rounds	-0.080	-0.201	-0.213***	-0.238***
	(0.082)	(0.127)	(0.034)	(0.065)
Industry × state FE	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes

Notes: Entries are $EU \times Data \times Policy$ coefficients. Each investor pair omits and includes investor-by-quarter effects, respectively. Both retain investor-by-industry-by-state and state-by-quarter effects. Panel B retains industry-by-state and state-by-quarter effects and has one estimate per policy/sample combination. UK investors and full-sample exposure are retained when UK ventures are excluded; continuing activity is reassessed outside the UK. Standard errors are two-way clustered by venture industry and state, with positive-semidefinite correction where required. Stars denote significance at the 1, 5, and 10 percent levels. Table B.6 reports retained samples.

Table B.6. Retained samples for the UK comparison

Panel A: US investor participations

	Observations in estimation				Investors in estimation			
	Full		No UK		Full		No UK	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All participation	1,865,828	1,057,574	1,817,468	1,022,569	12,155	7,883	12,049	7,779
Syndicate lead	466,574	204,694	452,990	196,090	4,734	2,621	4,667	2,581
Solo investor	473,554	193,310	460,402	186,312	4,732	2,391	4,664	2,345
Syndicate non-lead	1,262,510	627,741	1,232,534	607,686	10,488	6,461	10,387	6,374
Active from 2016Q2	1,718,992	1,050,895	1,671,857	1,015,993	7,516	7,131	7,445	7,032
Active from 2018Q2	1,559,708	1,015,414	1,514,425	980,805	5,668	5,518	5,616	5,447
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes

Panel B: US-only financing rounds

	Full	No UK
Observations in estimation	39,437	38,645
Industry × state FE	Yes	Yes
State × quarter FE	Yes	Yes

Notes: Counts refer to the observations and distinct investors retained by each Poisson model in Table B.5. Investor pairs omit and include investor-by-quarter effects. The industry outcome counts financing rounds and has no investor-level sample count.

Appendix C. Financing Amounts

The following tables report financing amounts in millions of US dollars for the outcomes analyzed in the count tables. Industry–state outcomes sum disclosed round amounts within each cell. Investor outcomes divide each disclosed round equally across its reconstructed participation records before aggregation, and exclude cells containing any undisclosed transaction. The resulting amounts proxy for financing associated with participation, not observed investor-specific check sizes. Comparisons with counts may reflect financing scale, disclosure, and changes in the estimation sample. The industry tables retain both specifications reported in Table 2; the investor tables retain only the paired specifications used in the main text. No dollar-amount event studies are presented.

Table C.1. Disclosed financing amounts and industry exposure

	Industry, state, and quarter FE					Industry × state, state × quarter FE				
	(1) Cross-union particip.	(2) Cross-union led	(3) Same-union led	(4) EU-only	(5) US-only	(6) Cross-union particip.	(7) Cross-union led	(8) Same-union led	(9) EU-only	(10) US-only
EU × *Data*	0.737	1.267**	0.537*	0.970***	1.367***	–	–	–	–	–
	(0.508)	(0.548)	(0.291)	(0.313)	(0.323)					
EU × *Enactment*	-0.426	-0.796*	0.241	-0.421***	0.429	–	–	–	–	–
	(0.268)	(0.441)	(0.153)	(0.060)	(0.367)					
EU × *Rollout*	-0.252	-0.511	0.017	-0.802	0.396	–	–	–	–	–
	(0.343)	(0.477)	(0.193)	(0.544)	(0.359)					
Data × *Enactment*	0.168	0.281	-0.143**	-0.274***	-0.248***	-0.046	-0.142	-0.175***	-0.319**	-0.295***
	(0.271)	(0.601)	(0.056)	(0.093)	(0.089)	(0.131)	(0.369)	(0.034)	(0.146)	(0.101)
Data × *Rollout*	-0.160	0.529	-0.399**	-0.716	-0.457**	-0.329**	0.239	-0.346***	-0.624	-0.394***
	(0.221)	(0.601)	(0.172)	(0.620)	(0.188)	(0.134)	(0.775)	(0.090)	(0.746)	(0.098)
EU × *Data* × *Enactment*	-0.311	-0.836	0.285	0.511**	-0.297	-0.232**	-0.622**	0.360	0.624***	-0.450
	(0.261)	(0.559)	(0.270)	(0.216)	(0.257)	(0.117)	(0.272)	(0.234)	(0.141)	(0.275)
EU × *Data* × *Rollout*	-0.740*	-1.615***	0.263	0.680	-0.964*	-0.652*	-1.585**	0.261	0.652	-1.289***
	(0.393)	(0.552)	(0.314)	(0.623)	(0.505)	(0.373)	(0.732)	(0.245)	(0.733)	(0.397)
Industry FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
State FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Quarter FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Industry × state FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
State × quarter FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Observations before pruning	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000
Observations in estimation	66,576	65,664	72,000	63,648	72,000	18,429	13,528	42,638	19,321	32,891

Notes: Outcomes sum disclosed round amounts in millions of US dollars and may overlap. Cross-union leadership requires at least one opposite-union lead; same-union leadership requires confirmed same-union-only leads. EU-only and US-only exclude investors from the opposite union but allow third-country or unlocated investors. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Dashes mark absorbed terms or excluded fixed effects. Positive-semidefinite covariance correction applies in columns 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Table C.2. Disclosed financing amounts and continuous exposure

	Industry, state, and quarter FE					Industry × state, state × quarter FE				
	(1) Cross-union particip.	(2) Cross-union led	(3) Same-union led	(4) EU-only	(5) US-only	(6) Cross-union particip.	(7) Cross-union led	(8) Same-union led	(9) EU-only	(10) US-only
EU × *Data*	0.302	0.656**	0.082	0.473**	0.572**	–	–	–	–	–
	(0.280)	(0.302)	(0.227)	(0.219)	(0.237)					
EU × *Enactment*	-0.534**	-0.992**	0.228*	-0.354***	0.356	–	–	–	–	–
	(0.257)	(0.461)	(0.130)	(0.074)	(0.384)					
EU × *Rollout*	-0.416	-0.853	0.004	-0.745	0.165	–	–	–	–	–
	(0.310)	(0.529)	(0.170)	(0.477)	(0.387)					
Data × *Enactment*	0.135	0.341	-0.086	0.177*	-0.141**	0.027	0.123	-0.110**	0.288	-0.179**
	(0.209)	(0.395)	(0.055)	(0.091)	(0.068)	(0.123)	(0.261)	(0.047)	(0.200)	(0.073)
Data × *Rollout*	0.010	0.371	-0.163	-0.050	-0.224	-0.084	0.142	-0.096	-0.161	-0.148**
	(0.157)	(0.281)	(0.146)	(0.268)	(0.169)	(0.125)	(0.322)	(0.065)	(0.400)	(0.073)
EU × *Data* × *Enactment*	-0.057	-0.436	0.272	0.007	-0.214	-0.003	-0.331	0.291	-0.064	-0.204
	(0.170)	(0.341)	(0.249)	(0.240)	(0.215)	(0.092)	(0.258)	(0.180)	(0.221)	(0.159)
EU × *Data* × *Rollout*	-0.284	-0.670***	0.181	0.083	-0.420	-0.208	-0.595*	0.149	0.235	-0.471**
	(0.246)	(0.255)	(0.257)	(0.291)	(0.307)	(0.251)	(0.359)	(0.170)	(0.390)	(0.229)
Industry FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
State FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Quarter FE	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Industry × state FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
State × quarter FE	–	–	–	–	–	Yes	Yes	Yes	Yes	Yes
Macroeconomic controls	Yes	Yes	Yes	Yes	Yes	–	–	–	–	–
Observations before pruning	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000	72,000
Observations in estimation	66,576	65,664	72,000	63,648	72,000	18,429	13,528	42,638	19,321	32,891

Notes: Outcomes sum disclosed round amounts in millions of US dollars and may overlap. Cross-union leadership requires at least one opposite-union lead; same-union leadership requires confirmed same-union-only leads. EU-only and US-only exclude investors from the opposite union but allow third-country or unlocated investors. *Data* is the pre-enactment data-related share standardized across the 40 classified industries; coefficients represent a one-standard-deviation contrast. Estimates are from Poisson regressions; policy definitions and industry/state clustering follow Section 3. Stars denote significance at the 1, 5, and 10 percent levels. Dashes mark absorbed terms or excluded fixed effects. Positive-semidefinite covariance correction applies in columns 1, 2, 3, 4, 5, 6, 7, 9, 10.

TABLE C.3. Allocated financing amounts by investor location

	US		EU	
	(1)	(2)	(3)	(4)
Data × *Enactment*	-0.173**	-0.326***	-0.159	-0.598**
	(0.071)	(0.064)	(0.223)	(0.253)
Data × *Rollout*	-0.305***	-0.586***	0.017	-0.471***
	(0.118)	(0.110)	(0.534)	(0.098)
EU × *Data* × *Enactment*	-0.439**	-0.170	0.147	0.457
	(0.179)	(0.233)	(0.267)	(0.328)
EU × *Data* × *Rollout*	-0.999***	-0.628***	-0.333	0.084
	(0.280)	(0.213)	(0.563)	(0.186)
Investor × industry × state FE	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–
Observations before pruning	1,846,100	1,846,100	755,129	755,129
Observations in estimation	1,569,573	854,638	575,267	263,700
Investors in estimation	11,355	7,167	5,577	3,341

Notes: Samples follow Table 6. Outcomes are allocated financing in millions of US dollars, constructed as described above; cells containing undisclosed rounds are excluded. Paired columns omit and include investor-by-quarter effects, respectively, and may retain different samples. Estimates are from Poisson regressions with standard errors clustered by industry and state. Stars denote significance at the 1, 5, and 10 percent levels. Positive-semidefinite covariance correction applies in column 4.

TABLE C.4. Allocated amounts by US investor role

	All		Syndicate lead		Solo		Syndicate non-lead	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Data × *Enactment*	-0.173**	-0.326***	-0.248***	-0.300**	-0.033	-0.613**	-0.154*	-0.330***
	(0.071)	(0.064)	(0.084)	(0.120)	(0.230)	(0.311)	(0.083)	(0.064)
Data × *Rollout*	-0.305***	-0.586***	-0.379***	-0.563***	-0.394	-1.262***	-0.277*	-0.594***
	(0.118)	(0.110)	(0.117)	(0.130)	(0.243)	(0.337)	(0.152)	(0.105)
EU × *Data* × *Enactment*	-0.439**	-0.170	-1.013***	-0.519***	-1.273	2.401*	-0.300*	0.352
	(0.179)	(0.233)	(0.353)	(0.066)	(1.290)	(1.278)	(0.175)	(0.415)
EU × *Data* × *Rollout*	-0.999***	-0.628***	-1.112***	-1.296**	-1.044	0.093	-0.952**	0.005
	(0.280)	(0.213)	(0.322)	(0.535)	(1.178)	(1.154)	(0.414)	(0.245)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–	–	–
Observations before pruning	1,846,100	1,846,100	1,846,100	1,846,100	1,846,100	1,846,100	1,846,100	1,846,100
Observations in estimation	1,569,573	854,638	404,875	171,826	251,841	76,589	1,174,531	571,612
Investors in estimation	11,355	7,167	4,327	2,324	3,591	1,557	10,110	6,139

Notes: Roles follow Table 7. Outcomes are allocated financing in millions of US dollars, constructed as described above; cells containing undisclosed rounds are excluded. Paired columns omit and include investor-by-quarter effects, respectively, and may retain different samples. Estimates are from Poisson regressions with standard errors clustered by industry and state. Stars denote significance at the 1, 5, and 10 percent levels. Positive-semidefinite covariance correction applies in columns 3, 4, 5.

Table C.5. Allocated amounts among continuing US investors

	All pre-active		Active from 2016Q2		Active from 2018Q2	
	(1)	(2)	(3)	(4)	(5)	(6)
Data × *Enactment*	-0.173**	-0.326***	-0.166***	-0.326***	-0.201***	-0.371***
	(0.071)	(0.064)	(0.064)	(0.064)	(0.060)	(0.060)
Data × *Rollout*	-0.305***	-0.586***	-0.300***	-0.586***	-0.293**	-0.607***
	(0.118)	(0.110)	(0.109)	(0.110)	(0.119)	(0.109)
EU × *Data* × *Enactment*	-0.439**	-0.170	-0.590**	-0.168	-0.389	-0.106
	(0.179)	(0.233)	(0.299)	(0.232)	(0.400)	(0.227)
EU × *Data* × *Rollout*	-0.999***	-0.628***	-1.156***	-0.626***	-0.935***	-0.600***
	(0.280)	(0.213)	(0.199)	(0.213)	(0.190)	(0.215)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–
Observations before pruning	1,846,100	1,846,100	1,700,337	1,700,337	1,542,659	1,542,659
Observations in estimation	1,569,573	854,638	1,448,048	849,845	1,318,752	824,010
Investors in estimation	11,355	7,167	7,349	6,582	5,583	5,221

Notes: Activity restrictions follow Table B.2. Outcomes are allocated financing in millions of US dollars, constructed as described above; cells containing undisclosed rounds are excluded. Continued activity can occur in either union and conditions on post-treatment behavior. Paired columns omit and include investor-by-quarter effects, respectively, and may retain different samples. Estimates are from Poisson regressions with standard errors clustered by industry and state. Stars denote significance at the 1, 5, and 10 percent levels.

Table C.6. Allocated amounts by US investor type

	Angel / accel.		VC		PE / growth		Corporate / CVC	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Data × *Enactment*	-0.429***	-0.541***	-0.268***	-0.400***	-0.222	-0.437*	-0.307**	-0.553***
	(0.056)	(0.128)	(0.067)	(0.085)	(0.163)	(0.262)	(0.123)	(0.169)
Data × *Rollout*	0.058	-0.492***	-0.413***	-0.655***	-0.723***	-0.979***	-0.023	-0.763***
	(0.257)	(0.139)	(0.100)	(0.134)	(0.164)	(0.197)	(0.387)	(0.173)
EU × *Data* × *Enactment*	0.139	0.037	-0.261	0.293	-1.466**	-0.469	1.202***	0.962***
	(0.474)	(0.246)	(0.196)	(0.290)	(0.594)	(0.378)	(0.149)	(0.164)
EU × *Data* × *Rollout*	-0.445	0.325	-0.805**	-0.280	-1.218***	-0.161	-0.437	0.687***
	(0.500)	(0.214)	(0.313)	(0.196)	(0.333)	(0.186)	(0.411)	(0.256)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–	–	–
Observations before pruning	637,037	637,037	824,235	824,235	121,087	121,087	138,783	138,783
Observations in estimation	480,641	198,508	744,127	496,019	97,094	46,606	117,800	54,619
Investors in estimation	4,877	2,753	3,236	2,584	865	563	1,319	633

Notes: Investor types follow Table B.3. Outcomes are allocated financing in millions of US dollars, constructed as described above; cells containing undisclosed rounds are excluded. Paired columns omit and include investor-by-quarter effects, respectively, and may retain different samples. Estimates are from Poisson regressions with standard errors clustered by industry and state. Stars denote significance at the 1, 5, and 10 percent levels. Positive-semidefinite covariance correction applies in columns 1, 2, 6, 8.

Table C.7. Allocated amounts by investment relationship

	All		Initial		Follow-on		Second		Third or later	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Data × *Enactment*	-0.173**	-0.326***	-0.163**	-0.327***	-0.162	-0.380***	-0.265**	-0.591***	-0.080*	-0.209
	(0.071)	(0.064)	(0.079)	(0.088)	(0.121)	(0.073)	(0.128)	(0.087)	(0.041)	(0.162)
Data × *Rollout*	-0.305***	-0.586***	-0.364***	-0.581***	-0.376***	-0.645***	-0.471***	-0.745***	-0.360**	-0.521***
	(0.118)	(0.110)	(0.128)	(0.096)	(0.127)	(0.157)	(0.133)	(0.204)	(0.169)	(0.108)
EU × *Data* × *Enactment*	-0.439**	-0.170	-0.532***	-0.244	-0.658***	0.073	-0.259*	-0.099	0.236	0.565
	(0.179)	(0.233)	(0.170)	(0.206)	(0.209)	(0.503)	(0.149)	(0.455)	(0.611)	(0.412)
EU × *Data* × *Rollout*	-0.999***	-0.628***	-0.887***	-0.674**	-1.085***	-0.414	-1.383**	-0.698	0.077	-0.691*
	(0.280)	(0.213)	(0.306)	(0.278)	(0.209)	(0.345)	(0.615)	(0.456)	(0.634)	(0.362)
Investor × industry × state FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Investor × quarter FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Macroeconomic controls	–	–	–	–	–	–	–	–	–	–
Observations before pruning	1,846,100	1,846,100	1,845,915	1,845,915	1,845,915	1,845,915	1,845,005	1,845,005	1,845,005	1,845,005
Observations in estimation	1,569,573	854,638	1,465,383	729,699	386,569	177,932	339,003	132,771	122,186	46,320
Investors in estimation	11,355	7,167	10,961	6,847	4,293	2,212	3,987	1,989	1,782	817

Notes: Relationship status and order follow Table B.4. Outcomes are allocated financing in millions of US dollars, constructed as described above; cells containing undisclosed rounds are excluded. Paired columns omit and include investor-by-quarter effects, respectively, and may retain different samples. Estimates are from Poisson regressions with standard errors clustered by industry and state. Stars denote significance at the 1, 5, and 10 percent levels. Positive-semidefinite covariance correction applies in columns 8, 9, 10.

Appendix D. Annual Dynamics and Pre-Enactment Diagnostics

The event studies estimate calendar-year interactions using quarterly observations and the fixed effects specified below. Annual interactions apply to $EU \times Data$, EU, and $Data$, with absorbed terms omitted and 2015 as the reference. Industry models include industry-by-state and state-by-quarter effects. Investor models include investor-by-industry-by-state, state-by-quarter, and investor-by-quarter effects. We estimate the annual coefficients directly rather than average quarterly estimates.

Annual interactions summarize changes across years but do not distinguish changes within the year of enactment or rollout. The 2016 coefficient includes quarters before and after April enactment, and 2018 spans May rollout. They cannot isolate immediate policy-date changes. Only 2014 precedes the reference, and only 2019 is a full post-rollout calendar year, limiting the assessment of pretrends and longer-run persistence.

Table D.1 tests whether each 2014 coefficient equals zero relative to 2015. The reported F tests use the clustered covariance matrix and the available cluster degrees of freedom; each has one numerator degree of freedom. Non-rejection is not proof of parallel trends. In particular, the continuous-exposure participation and leadership tests reject even though their binary counterparts do not. Figures use pointwise intervals of the estimate plus or minus 1.96 standard errors, rather than simultaneous bands.

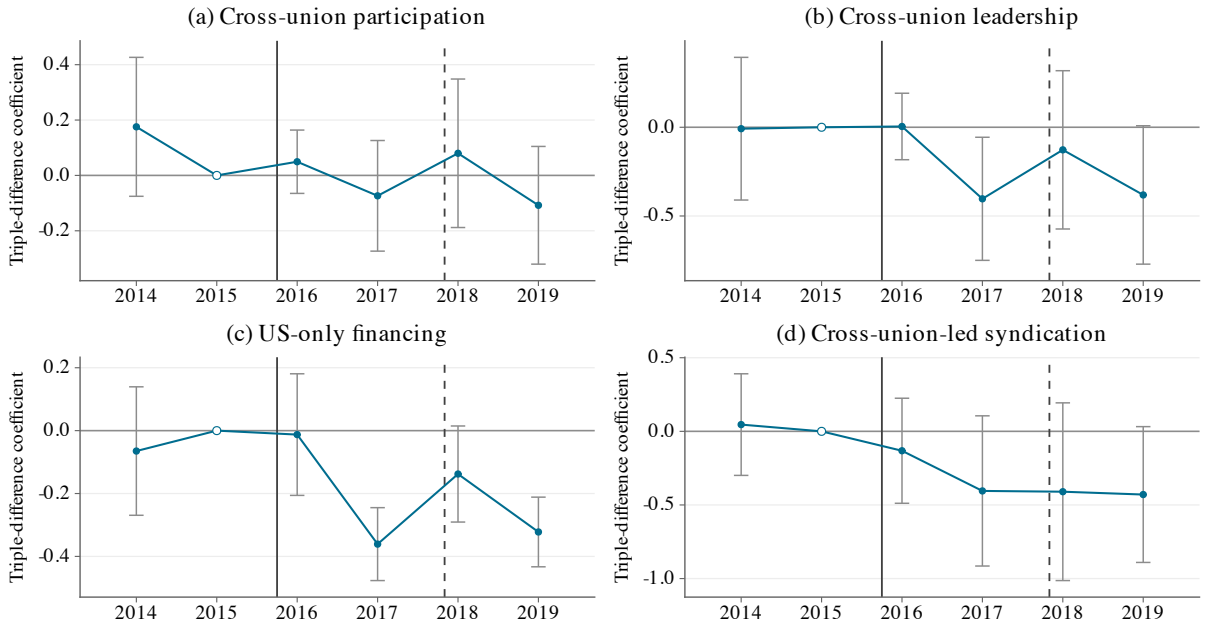
TABLE D.1. Annual pre-period tests

Outcome	F statistic	Denom. df	p-value
Figure 1			
All US participation	0.199	39	0.6581
Syndicate lead	0.046	38	0.8308
Solo investor	1.735	38	0.1957
Figure D.1			
Cross-union participation	1.879	37	0.1787
Cross-union leadership	0.002	37	0.9687
US-only financing	0.388	39	0.5368
Cross-union-led syndicated rounds	0.068	36	0.7958
Figure D.2			
Distance: all investors	0.016	39	0.8990
Distance: leads	0.612	39	0.4388
Figure D.3			
Seed and Startup: cross-union participation	0.994	35	0.3255
Seed and Startup: cross-union leadership	0.003	34	0.9582
Seed and Startup: US-only financing	0.501	39	0.4831
Figure D.4			
Continuous exposure: cross-union participation	5.280	37	0.0273
Continuous exposure: cross-union leadership	5.239	37	0.0279
Continuous exposure: US-only financing	0.042	39	0.8379
US-only financing (excluding UK)	0.060	39	0.8071
Figure D.5			
Angels and accelerators	1.175	39	0.2850

Notes: Each test concerns the 2014 $EU \times Data$ coefficient relative to the omitted 2015 coefficient and has one numerator degree of freedom. Models retain quarterly observations and industry-by-state and state-by-quarter effects for industry outcomes, and investor-by-industry-by-state, state-by-quarter, and investor-by-quarter effects for investor outcomes. Tests use two-way clustered covariance matrices with positive-semidefinite correction. These tests concern one pre-reference annual difference and cannot establish the absence of differential quarterly trends. Only displayed, nonduplicate annual paths are listed, grouped by the figure that displays them.

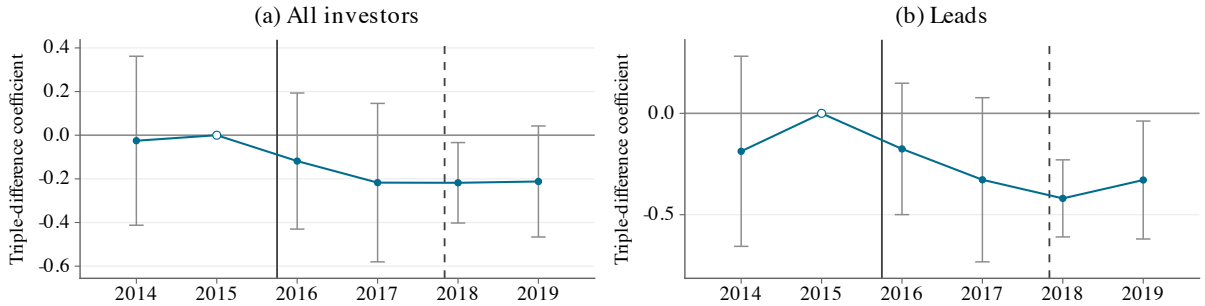
The displayed paths are drawn from count and distance outcomes whose static rollout triple interaction is significant at the 10 percent level in the preferred specification. Figure 1 presents the central investor paths in the main text. Non-lead participation has no available path under this rule, but remains in the main static table. Continuing-investor results also remain in the static tables. Selection of these paths does not establish identification.

Figure D.1. Annual dynamics of financing and syndicate leadership



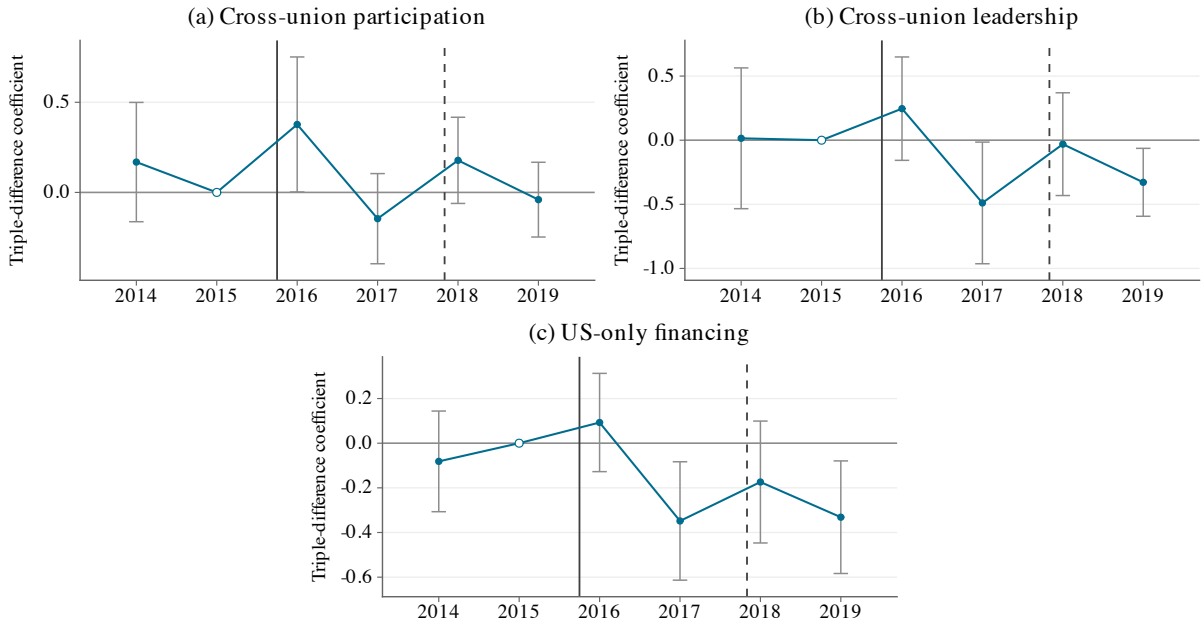
Notes: Panels (a)–(c) use industry-by-state and state-by-quarter effects, as in columns 6–10 of Table 2; panel (d) counts cross-union-led syndicated rounds in Table 3. These categories overlap. The models retain quarterly observations and impose common event coefficients within each calendar year, with 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout within the calendar years. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with standard errors clustered by industry and state and corrected for a non-positive-semidefinite covariance matrix where required. Table D.1 reports the annual pre-period tests.

Figure D.2. Annual dynamics of investor–venture distance



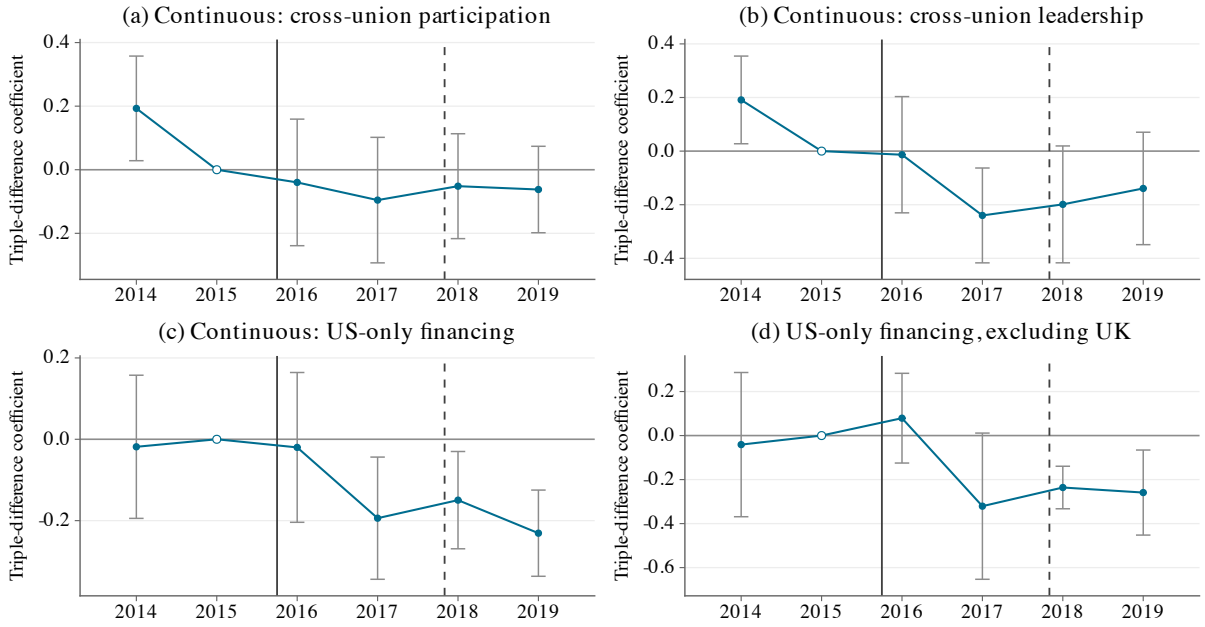
Notes: The outcomes are complete cell-average distances in miles for all EU/US investors and for leads, matching Table 4. Distances are conditional on completed investments. The models retain quarterly observations and impose common event coefficients within each calendar year, with 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout within the calendar years. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with standard errors clustered by industry and state and corrected for a non-positive-semidefinite covariance matrix where required. Table D.1 reports the annual pre-period tests.

FIGURE D.3. Annual dynamics of Seed and Startup financing



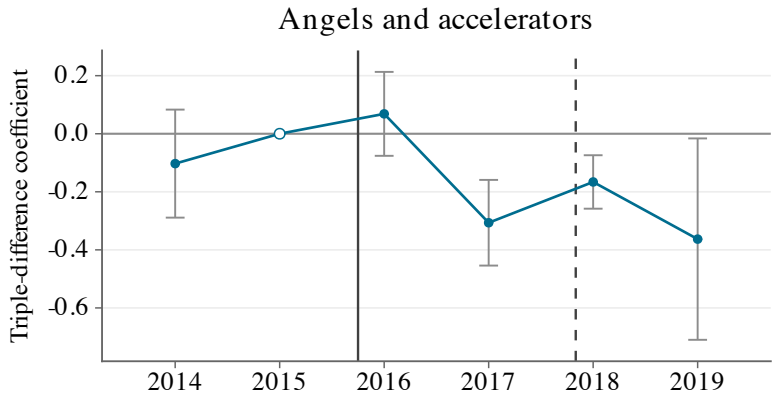
Notes: Panels (a)–(c) show Seed and Startup cross-union participation, cross-union leadership, and US-only financing, matching column 1 of Panels A–C in Table 5. Seed and Startup pools Accelerator/Incubator, Angel, and Seed rounds. The models retain quarterly observations and industry-by-state and state-by-quarter effects, with common event coefficients within each calendar year and 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with industry/state clustering and positive-semidefinite covariance correction. Table D.1 reports the annual pre-period tests.

FIGURE D.4. Annual dynamics under continuous exposure and UK exclusion



Notes: Panels (a)–(c) replace binary exposure with the standardized pre-enactment industry share, with industry-by-state and state-by-quarter effects. Panel (d) excludes UK ventures from the US-only count while retaining full-sample exposure. The first two continuous-exposure pre-period tests reject at the 5 percent level. The models retain quarterly observations and impose common event coefficients within each calendar year, with 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout within the calendar years. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with standard errors clustered by industry and state and corrected for a non-positive-semidefinite covariance matrix where required. Table D.1 reports the annual pre-period tests.

FIGURE D.5. Annual dynamics of angel and accelerator participation



Notes: The sample consists of pre-enactment US angel and accelerator investors. The model includes investor-by-industry-by-state, state-by-quarter, and investor-by-quarter effects. The models retain quarterly observations and impose common event coefficients within each calendar year, with 2015 omitted (open circle). Solid and dashed lines mark April 2016 enactment and May 2018 rollout within the calendar years. The 2016 and 2018 coefficients span policy changes. Bars are pointwise 95 percent confidence intervals with standard errors clustered by industry and state and corrected for a non-positive-semidefinite covariance matrix where required. Table D.1 reports the annual pre-period tests.